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Merit Aid, Sorting, and Institutional Access: Evidence from Tennessee's HOPE Scholarship*

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Abstract

Merit-based scholarship programs are widely viewed as regressive because eligibility correlates with family income. We show that this conclusion conflates individual attendance effects with institutional composition effects—distinct estimands that can move in opposite directions when students sort across multiple sectors. Using tax-linked data from Chetty et al. (2020), we find that Tennessee's HOPE Scholarship reshapes the full parental income distribution at public colleges: all five income quintiles shift monotonically, with bottom-quintile share rising and top-quintile share falling. Total enrollment is unchanged; the shift reflects three-destination sorting in which lower-income students enter from non-attendance while upper-income students exit to out-of-state alternatives. A permutation test ($p = 0.037$ for parental income rank, $p = 0.074$ for bottom-quintile share) and synthetic control analysis confirm Tennessee as an outlier. Results are robust across four heterogeneity-corrected estimators. Massachusetts' Adams Scholarship independently produces a progressive shift of similar magnitude under similar design parameters, while South Dakota's high-threshold program produces the opposite pattern—as the model predicts. A three-destination sorting framework reconciles these progressive composition effects with the regressive attendance findings of Dynarski (2000) and identifies program design parameters that determine sorting direction.

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1 Introduction

Dynarski (2000) showed that Georgia's HOPE Scholarship widened the gap in college attendance between higher- and lower-income students. The finding crystallized a distributional critique of merit aid: because eligibility is conditioned on academic achievement, which correlates with family income, broad-based merit programs disproportionately subsidize students who would attend college regardless (Dynarski, 2004; Jones et al., 2022). This view now dominates the policy debate.

A systematic review of 71 quasi-experimental studies finds that merit-based grants rarely improve outcomes for disadvantaged students and can increase inequality (Herbaut & Geven, 2020). We document an exception.

Using tax-linked income data from Chetty et al. (2020), we estimate how Tennessee's HOPE Scholarship reshaped the full parental income distribution at public colleges. The results are the opposite of what the conventional view predicts: following HOPE implementation, public colleges experienced a *progressive* compositional shift. Bottom-quintile students gained share at public institutions, while top-quintile students lost share. The effect is monotonic across all five income quintiles—Q1 through Q3 gain, Q4 and Q5 lose—and every coefficient is statistically significant. This monotonic gradient is the paper's central empirical result.

Aggregate attendance effects and institutional composition effects are distinct estimands that need not move in the same direction. In a two-sector model—attend college or not—individual enrollment responses fully determine institutional composition. But when students choose among public, private, and out-of-state alternatives, the link breaks: the *aggregate* attendance gap can widen (because higher-income students are more likely to qualify) while the *sector-specific* composition of public colleges simultaneously improves (because the students who actually sort into the public sector are drawn disproportionately from the bottom of the income distribution).[1] The reconciliation depends on where displaced students go. If top-quintile students exit to out-of-state or private alternatives rather than entering public colleges, the public sector becomes more progressive even as overall college-going becomes more unequal. We formalize this insight in a three-destination sorting model (public, private, out-of-state) and test its predictions directly.

[1] This is a form of Simpson's paradox: the aggregate relationship between merit aid and income composition reverses sign when conditioned on enrollment sector. The aggregate effect reflects who attends college; the composition effect reflects who attends *which* college.

Institutional composition matters independently of individual enrollment decisions. The peer effects literature demonstrates that student body composition shapes academic outcomes (Sacerdote, 2001). Chetty et al. (2020) show that access to high-mobility institutions drives intergenerational economic outcomes, establishing that who attends which institution has first-order welfare consequences. If merit aid systematically alters the income profile of public college cohorts, it reshapes the peer environment and mobility prospects for all enrolled students—not only those on the margin of attendance.

Four pieces of evidence are consistent with a sorting interpretation. First, total enrollment at Tennessee public colleges is unchanged after HOPE. The share shifts reflect real reallocation—11 additional bottom-quintile students per cohort enter, 34 top-quintile students per cohort exit—not a denominator artifact. Second, Tennessee private colleges show no mirror image of the public-sector shift: the decline in top-quintile representation at public colleges is not offset by gains at Tennessee privates, consistent with out-of-state exit as the relevant margin. Third, two-year and four-year institutions adjust through different channels. Community colleges exhibit displacement (top-quintile students replaced by bottom-quintile students, total seats unchanged), while four-year colleges show expansion (enrollment grows, composition shifts). Fourth, within-state comparisons restricted to Tennessee institutions confirm that the composition change is specific to the public sector.

These patterns may not be idiosyncratic to Tennessee. Our theoretical framework proposes that program design and institutional context jointly shape the direction of compositional sorting, through three parameters: the eligibility threshold relative to the income distribution, the size of the public-private subsidy differential, and interactions with existing need-based aid. Georgia's HOPE, which initially offset Pell Grant awards dollar-for-dollar and imposed an income cap, eliminated the access margin for the lowest-income families and muted the public-private subsidy differential—conditions under which the model predicts regressive sorting. Dynarski's findings for Georgia are consistent with this prediction, as are the enrollment patterns documented by Cornwell et al. (2006). Tennessee's HOPE, which has no Pell offset, no income cap, and a large public-private subsidy differential, satisfies the conditions for progressive sorting. The same framework explains both outcomes.

We test these predictions out of sample. Massachusetts' Adams Scholarship (Cohodes & Good-

man, 2014), which shares Tennessee's design features (moderate threshold, public-sector subsidy concentration), independently produces a progressive shift of similar magnitude (+1.38 pp for bottom-quintile share). South Dakota's Opportunity Scholarship, which imposes a high ACT threshold and small award, produces a strongly regressive shift—as the model predicts. West Virginia's PROMISE, despite similar design parameters to Tennessee, shows no compositional effect, highlighting that program design is necessary but not sufficient: enrollment contraction at WV public colleges overwhelmed any sorting dynamics.

We estimate treatment effects using Callaway and Sant'Anna (2021) as the primary estimator, with TWFE, Sun and Abraham (2021), and de Chaisemartin and D'Haultfœuille (2020) for robustness. A bordering-state control group passes pre-trend tests that the national comparison fails at longer horizons. Results are stable across three treatment timing definitions and two alternative control group specifications. Under the relative-magnitude restriction of Rambachan and Roth (2023), the estimated effect survives violations up to twice the magnitude observed in the pre-treatment period. Under the stricter smoothness restriction, results are more fragile. We interpret the evidence as credible but not definitive, and report all sensitivity analyses transparently.

Our work connects three literatures. First, a large body of research documents the enrollment effects of merit aid (Bruce & Carruthers, 2014; Cohodes & Goodman, 2014; Cummings et al., 2022; Dynarski, 2000, 2004; Song & Rubenstein, 2024; Verzyl, 2023). This literature focuses on individual enrollment decisions, not institutional composition. Second, studies of policy-induced sorting show that admissions and financing reforms can substantially reshape who attends which institutions (Bleemer, 2022; Card & Krueger, 2005; Hinrichs, 2012). Third, Chetty et al. (2020) demonstrate that access to high-mobility institutions shapes intergenerational economic outcomes, establishing that institutional composition has first-order welfare consequences. We bridge these literatures by documenting how merit aid is associated with changes in the full income distribution across institutional sectors.

This paper makes three contributions. First, we provide the first evidence on how merit aid reshapes the *full income distribution* at public institutions, using tax-linked data that measures parental income across all quintiles rather than relying on Pell eligibility as a binary proxy. The monotonic quintile gradient—all five coefficients significant and perfectly ordered—has not been documented in any prior study of merit aid. Second, we develop a three-destination sorting frame-

work that reconciles the progressive institutional composition effects we find with the regressive attendance effects documented by Dynarski (2000), showing that the apparent contradiction dissolves once student destinations are disaggregated. Third, we propose that program design parameters—Pell interaction, subsidy differential, threshold location—shape whether merit aid is associated with progressive or regressive sorting, and test the framework across four states: Tennessee and Massachusetts (progressive, as predicted), South Dakota (regressive, as predicted), and West Virginia (null, confounded by enrollment contraction).

The paper proceeds as follows. Section 2 describes the Tennessee HOPE Scholarship and the Georgia program that serves as our theoretical comparison case. Section 3 presents the three-destination sorting model and derives testable predictions. Section 4 describes the data. Section 5 discusses the empirical strategy. Section 6 presents results. Section 7 reports robustness checks. Section 8 discusses cross-state evidence from Georgia, Massachusetts, South Dakota, and West Virginia. Section 9 concludes.

2 Institutional Background

2.1 Tennessee HOPE Scholarship

Tennessee introduced the HOPE Scholarship in 2003, funded by proceeds from the state education lottery (Tennessee Higher Education Commission, 2025). The program provides tuition and fee subsidies to Tennessee residents who graduate from an eligible high school with a minimum 3.0 GPA or score at least 21 on the ACT (or 980 on the SAT). Recipients must maintain a 2.75 cumulative GPA after the first semester to retain eligibility (Ness & Tucker, 2008).

The scholarship provides a fixed dollar award covering approximately 60–80% of in-state tuition at public institutions governed by the Tennessee Board of Regents (TBR) and the University of Tennessee system ($3,000 for freshmen and sophomores, scaling to $6,000 for juniors and seniors at four-year institutions during the study period). Students attending eligible private institutions receive a smaller fixed amount ($3,000–$4,000 per year), which covers a substantially lower share of private tuition. This differential generosity creates a relative price advantage for public colleges—the key mechanism in our theoretical framework.

Table A1 reports the distribution of HOPE recipients across sectors for the first five entering

cohorts (2004–2008). Public institutions—TBR universities, TBR community colleges, and UT campuses—account for approximately 83–85% of recipients in every year. Private colleges receive 15–17%. This concentration of aid in the public sector motivates our identification strategy: we treat Tennessee public colleges as the primary recipients of the HOPE demand shock.

2.2 Prior Research on TN HOPE

Several studies examine Tennessee's merit aid program. The closest antecedent is Bruce and Carruthers (2014), who use a regression discontinuity at the ACT score cutoff for HOPE eligibility. Chakrabarti and Roy (2022) study merit aid and student mobility more broadly, documenting how scholarship programs interact with college selectivity. Bruce and Carruthers (2014) find that Tennessee HOPE had no effect on overall enrollment among Pell-eligible students near the ACT threshold, but reallocated them from two-year to four-year public institutions (-2.9 percentage points at community colleges, $+2.4$ at four-year publics). For non-white students, overall enrollment declined by 2.6 percentage points.

Our analysis is complementary but asks a different question. Bruce and Carruthers estimate the *individual enrollment response* at the eligibility margin: how does a student just above the ACT cutoff choose differently from one just below? Their RD is local to the threshold and identifies the effect for compliers at that margin. We estimate the *institutional composition effect* across the full income distribution: how does the parental income profile of the entire student body at public colleges change after HOPE? Our difference-in-differences design captures the average treatment effect across all treated cohorts, including students well above the eligibility threshold for whom the RD has no power. The distinction matters because the composition of public colleges depends on the sorting behavior of students at all income levels, not only those at the eligibility margin. Our quintile decomposition—showing that all five quintiles shift monotonically—documents a pattern that no threshold-local design can detect, since students in the top quintile are almost all well above any reasonable ACT cutoff. Their finding that lower-income marginal students upgrade from community colleges to four-year institutions is consistent with our result that bottom-quintile share rises and that two-year and four-year colleges adjust through different channels.

2.3 Timeline and Cohort Mapping

The first HOPE recipients entered college in Fall 2003. In the Chetty et al. (2020) data, students are assigned to institutions attended between ages 19 and 22, and cohorts are indexed by birth year. The 1984 birth cohort (entering college in Fall 2002) is the last fully untreated cohort. The 1985 cohort is partially contaminated by a bridge provision that extended limited benefits to students who narrowly missed the first eligibility window. Our preferred treatment timing is $t^* = 1986$ (Fall 2004 entrants), the first cohort with unambiguous full exposure. We test robustness to $t^* \in \{1985, 1986, 1987\}$ in Section 7.

The analysis window spans birth cohorts 1980–1991, providing six pre-treatment cohorts (1980–1985) and six post-treatment cohorts (1986–1991) under the preferred timing. Table 1 maps each birth cohort to its event time, approximate college entry year, and HOPE exposure status. Each cohort's composition is measured independently: the outcome for cohort t at college c reflects the parental income distribution of students born in year t who attended college c between ages 19 and 22.

Table 1: Cohort Mapping: Birth Year, Event Time, and HOPE Exposure

Birth Cohort	Event Time (k)	College Entry	HOPE Status
1980	-6	Fall 1998	No HOPE
1981	-5	Fall 1999	No HOPE
1982	-4	Fall 2000	No HOPE
1983	-3	Fall 2001	No HOPE
1984	-2	Fall 2002	No HOPE (last untreated)
1985	-1	Fall 2003	Bridge provision (partial)
1986	0	Fall 2004	Full HOPE (t^*)
1987	$+1$	Fall 2005	Full HOPE
1988	$+2$	Fall 2006	Full HOPE
1989	$+3$	Fall 2007	Full HOPE
1990	$+4$	Fall 2008	Full HOPE
1991	$+5$	Fall 2009	Full HOPE

Notes: College entry is approximate (birth year + 18). Students are assigned to institutions attended between ages 19 and 22 in the Chetty et al. (2020) data. Each cohort's parental income composition is measured independently. The 1985 cohort received partial benefits through a bridge provision (Section 2).

Several other policy changes coincided with the HOPE Scholarship's introduction. Tennessee's education lottery launched in 2003 and generated revenue that funded both HOPE and Pre-K programs. The federal No Child Left Behind Act (2002) altered K–12 accountability requirements nationwide. Neither policy, however, would produce the specific pattern we document: a perfectly ordered income quintile gradient at public colleges with no effect at private institutions. A generic enrollment shock—whether driven by K–12 quality, state finances, or demographic change—would affect quintile shares unevenly but not produce five coefficients that are monotonically ordered and all statistically significant. The quintile gradient is a prediction unique to the price-reduction mechanism in the three-destination model, which makes it a discriminating test of the HOPE channel specifically.

2.4 Georgia HOPE as a Comparison Case

Georgia's HOPE Scholarship (1993) is the canonical case in the merit aid literature (Dynarski, 2000; Song & Rubenstein, 2024) and our primary theoretical comparison. Three design features distinguish it from Tennessee's program in ways that matter for our sorting framework.

First, from 1993 to 2000, Georgia HOPE awards were reduced dollar-for-dollar by Pell Grant amounts. Students from the lowest-income families received zero net benefit from HOPE for the program's first seven years. Dynarski (2000) documents the consequence: Black students, who were disproportionately Pell-eligible, showed no attendance gain, while white students gained 12.3 percentage points. The Pell offset effectively eliminated the access margin for the bottom of the income distribution.

Second, Georgia imposed an income cap ($66,000, later $100,000, eliminated in 1995) during the program's early years. While the cap was removed relatively quickly, its presence during the launch shaped initial enrollment patterns.

Third, Georgia HOPE provided comparable awards to private colleges: initially full tuition, later a fixed $3,000 that covered a meaningful share of private tuition at Georgia's price levels. This smaller public-private subsidy differential muted the sector-switching mechanism that drives compositional sorting in our model. Cornwell et al. (2006) confirm that four-year private enrollment in Georgia *rose* by 13.2% after HOPE, and two-thirds of the four-year enrollment gain reflected reduced out-of-state migration rather than new college-goers.

We return to the GA-TN comparison in Section 3.5, where we show that these design differences generate opposite predictions from the same theoretical framework.

3 Theoretical Framework

We develop a simple model of college choice that generates testable predictions about how merit aid reshapes the income composition of public institutions. The key departure from standard two-sector models is that we allow three destinations: in-state public college, in-state private college, and an outside option (out-of-state enrollment or non-attendance).

3.1 Setup

Consider a population of families indexed by income y_i and student ability a_i, where y_i and a_i are positively correlated. Each family chooses among three options: a public college (C_P), a private college (C_R), or no college (0). The indirect utility of family i from option $j \in \{P, R, 0\}$ is

$$V_{ij} = q_j - \frac{C_j}{y_i} + \varepsilon_{ij}, \tag{1}$$

where q_j is the quality of institution j, C_j is the sticker price, and ε_{ij} captures idiosyncratic preferences. We normalize $q_0 = C_0 = 0$. Assume $q_R > q_P > 0$ and $C_R > C_P > 0$: private colleges are both higher quality and more expensive.

3.2 Merit Scholarship Introduction

A state merit scholarship of amount $S > 0$ reduces the effective price of the public college for students with ability above a threshold τ:

$$C_P^{\text{eff}}(a_i) = \begin{cases} C_P - S & \text{if } a_i \geq \tau, \\ C_P & \text{if } a_i < \tau. \end{cases} \tag{2}$$

Without the scholarship, families sort across options by income. Define the no-college/public boundary as $\underline{y}$, where $V_{iP} = V_{i0}$, and the public/private boundary as $\overline{y}$, where $V_{iP} = V_{iR}$. In the absence of idiosyncratic preferences:

$$\underline{y} = \frac{C_P}{q_P}, \qquad \overline{y} = \frac{C_R - C_P}{q_R - q_P}. \tag{3}$$

Families with $y_i < \underline{y}$ do not attend college; those with $\underline{y} \leq y_i < \overline{y}$ attend public; those with $y_i \geq \overline{y}$ attend private.

The scholarship shifts both boundaries for eligible students:

$$\underline{y}^S = \frac{C_P - S}{q_P} < \underline{y}, \qquad \overline{y}^S = \overline{y} + \frac{S}{q_R - q_P} > \overline{y}. \tag{4}$$

The lower boundary falls (more families can afford public college) and the upper boundary rises

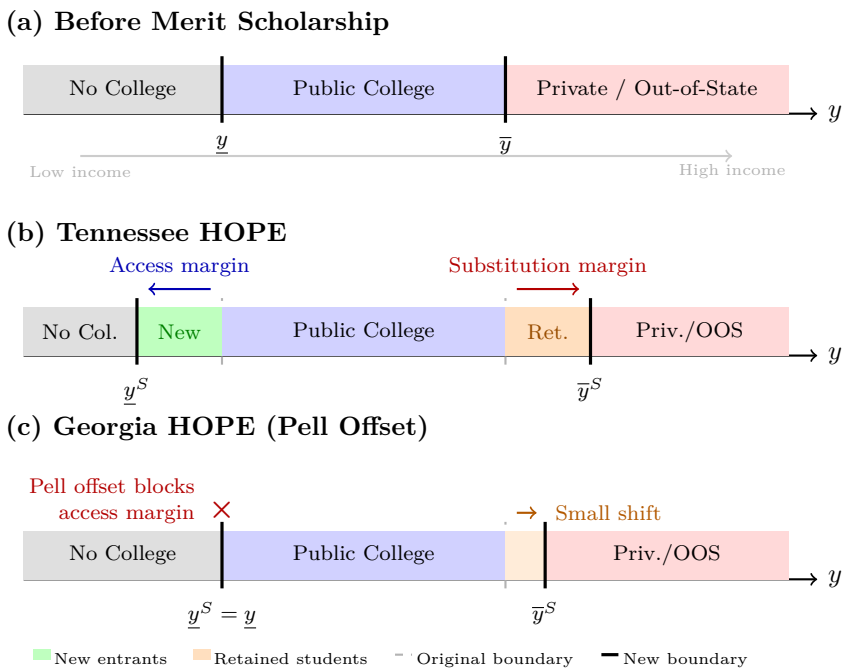

Figure 1: Three-Destination Sorting: How Merit Scholarships Reshape Institutional Composition

Notes: Panel (a) shows pre-scholarship sorting by family income y. Panel (b) shows Tennessee HOPE: the scholarship reduces the effective price of public college, shifting the lower boundary left (access margin: low-income students enter) and the upper boundary right (substitution margin: high-income students retained in public rather than exiting to private). The public sector expands in both directions, producing progressive sorting. Panel (c) shows Georgia HOPE: the Pell offset eliminates the effective subsidy for bottom-quintile families, so the lower boundary does not shift (×). Only the upper boundary moves modestly, producing regressive sorting.

(the public option becomes competitive for families who previously chose private). The scholarship expands the public college's enrollment base in both directions. Figure 1 illustrates the sorting mechanism.

3.3 Testable Predictions

The model generates seven predictions:

P1: Bottom-quintile share rises at public colleges. The scholarship reduces the effective price of public college. Because S/y_i is largest for low-income families, the marginal entrants drawn from below $\underline{y}$ are disproportionately low-income. We test P1 in Section 6.1.

P2: Monotonic quintile gradient. The price reduction S/y_i is monotonically decreasing in income. Q1 families experience the largest proportional price reduction; Q5 the smallest. The composition shift should be monotonically graded across quintiles. We test P2 in Section 6.3.

P3: Composition change reflects sorting, not enrollment expansion. Total enrollment at public colleges is unchanged if capacity constraints bind; the share shifts are driven by changes in who enrolls, not in how many enroll. We test P3 in Section 6.4.

P4: No mirror image at private colleges. If displaced top-quintile students exit to out-of-state options rather than shifting to private colleges, private institutions should show no significant composition change. We test P4 in Section 6.4.

P5: Heterogeneous channels by tier. At two-year colleges, where the no-college margin is active, the scholarship expands access (pulls in new students from below $\underline{y}$). At four-year universities, where the public/private margin is active, the scholarship induces sorting—a form of the ability stratification documented by Cornwell and Mustard (2006). We test P5 in Section 6.4.

P6: Spatial heterogeneity. Institutions located near state borders, where out-of-state alternatives are accessible, should show larger composition shifts because the outside option is more salient and the three-destination mechanism is most active. We test P6 in Section 6.5.

P7: Cross-state variation. Programs with no Pell offset and large public-private subsidy differentials produce progressive sorting. Programs with Pell offsets or comparable private awards produce regressive sorting or no composition change. The direction depends on whether the access margin operates for low-income students and whether $(S - s)$ is large enough to induce sector switching. We test P7 in Section 8.

3.4 Reconciling with Dynarski (2000)

Dynarski (2000) finds that Georgia's HOPE Scholarship widened the attendance gap between higher- and lower-income students. Our framework shows that this aggregate finding is consistent with a progressive composition shift at public institutions. Because eligibility is correlated with income, merit aid increases the *total* number of higher-income students attending college more than the total number of lower-income students. But the *destination* of these additional higher-income students matters. If they enroll at private or out-of-state institutions—or if they would have attended college anyway—the composition of public colleges can still shift downward.

The reconciliation rests on distinguishing between two quantities: the aggregate attendance gap (which can widen) and the institutional composition of the public sector (which can simultaneously become more progressive). Both are consistent with three-destination sorting.

This distinction is specific to the multi-sector setting. In a two-sector model—attend college or not—the aggregate attendance effect fully determines composition: if merit aid increases enrollment disproportionately among higher-income students, those students necessarily appear at colleges, and composition becomes more affluent. With three or more destinations, the link breaks. Higher-income students may attend college at higher rates but sort into private or out-of-state institutions, leaving the public sector's composition to be determined by the sorting behavior of remaining students. Individual enrollment responses are therefore insufficient to predict institutional composition; the full destination distribution is required. Our model captures this by allowing sorting across all three margins simultaneously.

3.5 Georgia as a Regressive Prediction

Georgia's HOPE included a Pell offset: scholarship dollars were reduced dollar-for-dollar by Pell Grant receipts (Figure 1, Panel C). Because Pell Grants target Q1 families, the effective scholarship for the poorest students was

$$S_{\text{eff}}^{Q1} = \max(S - \text{Pell},\ 0) \approx 0. \tag{5}$$

The model predicts no composition shift for Q1 (the price reduction is zero), while Q3–Q5 families receive the full S. The predicted effect is regressive: public colleges gain middle- and upper-income students without gaining low-income students. Dynarski (2000) documents the consequence: Black students, who were disproportionately Pell-eligible, showed no attendance gain, while white students gained 12.3 percentage points.

Georgia's HOPE also provided comparable awards to private colleges, muting the public-private subsidy differential. Cornwell et al. (2006) confirm the pattern: four-year private enrollment in Georgia *rose* by 13.2% after HOPE, and two-thirds of the four-year enrollment gain reflected reduced out-of-state migration rather than new college-goers.

Tennessee's HOPE has no Pell offset, no income cap, and a large public-private subsidy differential (full tuition at publics vs. \$3,000–\$4,000 at privates). Under these parameters, the model

predicts that the access margin and substitution margin both operate, producing the progressive gradient we observe. The TN-GA contrast is a pair of outcomes predicted by the same model under different parameter values.

4 Data

4.1 Chetty et al. (2020) Panel

We use the college-level panel constructed by Chetty et al. (2020), which links tax records and college enrollment data to measure the parental income distribution of students at each institution. The panel covers 2,202 colleges across 52 states and territories for birth cohorts 1980–1991 (approximately corresponding to college entry years 1998–2009). Students are assigned to the institution attended at ages 19–22.

The primary analysis table (MRC Table 3) provides college-by-cohort observations with measures of parental income rank and quintile shares. We supplement this with institutional characteristics from MRC Table 10 (SAT scores, expenditures, enrollment, Barron's selectivity, HBCU and flagship status, graduation rates). These characteristics capture the dimensions of institutional quality that prior work has linked to earnings premia (Brewer et al., 1999; Dale & Krueger, 2002).

4.2 Sample Construction

We define treated institutions as Tennessee public colleges. Tennessee has 51 colleges in the data, of which 26 are public—19 two-year institutions and 7 four-year institutions. Multi-campus systems appear as distinct identifiers (56 per cohort), but we identify 26 unique public colleges.

The control group comprises public colleges in states that did not adopt broad-based merit scholarship programs during the analysis period. We drop all 25 states classified as merit-program adopters by Sjoquist and Winters (2015). The remaining 26 control states contribute 6,289 college-cohort observations. Of the 312 possible Tennessee public-college-by-cohort observations (26×12), 291 are non-missing on both primary outcomes.

We exclude 24 observations with missing state or institution type.

4.3 Outcome Variables

Our two primary outcomes are:

- **Bottom-quintile share** (Q_1): the fraction of students at college c in cohort t whose parents are in the bottom 20% of the national income distribution. Baseline mean for Tennessee public colleges: 0.138.
- **Mean parental income rank** ($\overline{r}$): the average percentile rank of parental household income for students at college c in cohort t. Baseline mean for Tennessee publics: 0.540.

A clarification on the unit of observation: each outcome is measured at the college-by-birth-cohort level, not for the entire enrolled student body at a point in time. For example, Q_1 for the University of Tennessee in cohort 1986 measures the fraction of all 1986-born students who attended UT between ages 19 and 22 whose parents were in the bottom quintile. The Fall 2006 student body at UT, by contrast, would include birth cohorts 1984–1987 simultaneously—mixing untreated and treated cohorts. The cohort-specific measure provides sharper identification because it isolates students who faced HOPE-era incentives from those who did not. By the end of the post-treatment window ($k = +5$, cohort 1991), nearly all cohorts enrolled at a given institution are treated, so cohort-level and student-body-level effects converge.

For the quintile decomposition, we additionally use the share of students from each income quintile (Q_1 through Q_5).

For the level decomposition that addresses ecological inference concerns, we construct count outcomes by multiplying quintile shares by cohort size. For example, Q1 count $= Q_1 \times N_{ct}$, where N_{ct} is total enrollment at college c in cohort t.

4.4 Summary Statistics

Table 2 reports summary statistics for the primary outcomes and quintile shares, separately for Tennessee public colleges and control-state public colleges. Tennessee publics have a somewhat higher baseline share of bottom-quintile students (14.0% vs. 13.6%) and lower mean parental income rank (0.524 vs. 0.541). The quintile distribution at Tennessee publics is shifted leftward: Q1 and Q2 shares are higher, Q5 share is lower. Table A2 reports pre- and post-treatment means. After

HOPE implementation, bottom-quintile share at Tennessee publics rises to 15.9%, while the control group increases modestly to 12.9%. The raw difference-in-differences is 1.5 percentage points.

Table 2: Summary Statistics

	TN Public		Control Public	
	Mean	SD	Mean	SD
Primary Outcomes				
Bottom-quintile share (Q_1)	0.140	0.046	0.136	0.083
Mean parental income rank	0.524	0.067	0.541	0.103
Quintile Shares				
Q1 (bottom)	0.140		0.136	
Q2	0.195		0.182	
Q3	0.239		0.227	
Q4	0.256		0.252	
Q5 (top)	0.170		0.203	
Observations	137		2,850	
Colleges	26		∼500	

Notes: Pre-treatment means (birth cohorts 1980–1985) for Tennessee public colleges (treated) and public colleges in 26 non-merit-program states (control). Quintile shares refer to the fraction of students whose parents are in the indicated quintile of the national income distribution. Observations are at the college-by-cohort level. Data: Chetty et al. (2020).

5 Empirical Strategy

5.1 Primary Estimator: Callaway and Sant'Anna (2021)

Our primary estimator is the group-time average treatment effect on the treated (ATT) proposed by Callaway and Sant'Anna (2021). This estimator avoids the negative weighting and heterogeneity bias that can afflict two-way fixed effects specifications when treatment effects vary over time (Goodman-Bacon, 2021; Sun & Abraham, 2021).

In our setting, treatment timing is sharp: all Tennessee public colleges are treated simultaneously at $t^* = 1986$ (the first birth cohort with unambiguous HOPE exposure). The Callaway–Sant'Anna estimator computes group-time ATTs by comparing the outcome evolution of the treated group to a "never-treated" comparison group. We aggregate the group-time ATTs into an overall ATT and dynamic event-study coefficients.

We report TWFE event-study estimates for comparability with the prior difference-in-differences literature (Ashenfelter & Card, 1984; Card & Krueger, 1994). The TWFE specification is:

$$y_{ct} = \alpha_c + \lambda_t + \sum_{k=-5}^{5} \beta_k \cdot \mathbf{1}[t - t^* = k] \cdot \text{Treated}_c + \varepsilon_{ct} \tag{6}$$

where y_{ct} is the outcome for college c in birth cohort t, α_c are college fixed effects, λ_t are cohort fixed effects, Treated_c indicates Tennessee public colleges, and $k = -1$ is the omitted reference period.

5.2 Identification

Identification requires that Tennessee public colleges would have followed parallel trends in socioeconomic composition to control colleges absent the HOPE Scholarship. We assess this assumption in three ways.

First, we inspect pre-treatment event-study coefficients. Under the national control group, coefficients at $k = -5$ through $k = -3$ show some divergence from zero for bottom-quintile share, raising concerns about longer-horizon pre-trends. The $k = -5$ coefficient is driven by a single cohort (1980 births) and its magnitude is sensitive to the inclusion of this endpoint.

Second, we implement a bordering-state control group that restricts comparisons to colleges in states adjacent to Tennessee. This specification passes pre-trend tests: pre-treatment coefficients are small, statistically insignificant, and show no systematic pattern. The bordering-state specification is our preferred identification strategy.

Third, we implement a conservative control group that drops states with any policy changes in higher education funding during the analysis window. Results are quantitatively similar to the baseline.

5.3 Inference

Standard errors are clustered at the state level, the level at which treatment is assigned. With 27 clusters (26 control states plus Tennessee), we are near the boundary where cluster-robust inference becomes unreliable. In the Callaway–Sant'Anna framework, we use the analytical standard errors provided by the `csdid` package, which account for the two-step estimation.

For the bordering-state specification, which has only four state clusters ($G_1 = 1$ treated), the wild cluster bootstrap is uninformative (MacKinnon & Webb, 2017). We therefore supplement analytical standard errors with a Fisher-style permutation test that assigns placebo treatment to each state in turn and ranks Tennessee's estimate against the resulting distribution. We report permutation p-values in Section 7.5.

5.4 Sensitivity Analysis

We implement the honest confidence intervals of Rambachan and Roth (2023), which relax the parallel trends assumption by allowing for bounded violations. We report results under two restrictions:

- **Relative magnitude** (Δ_{RM}): post-treatment violations of parallel trends are bounded by a multiple $\bar{M}$ of the maximum pre-treatment violation. Results remain significant at $\bar{M} \leq 2$ for bottom-quintile share.
- **Smoothness** (Δ_{SD}): consecutive violations of parallel trends are bounded in their rate of change. Under this restriction, confidence intervals include zero at moderate bounds, indicating fragility. We interpret this as a limitation: if differential trends can accelerate smoothly, our estimates cannot be distinguished from pre-existing dynamics.

We discuss these results in detail in Section 7 and interpret the tension between Δ_{RM} and Δ_{SD} as informative about the nature of potential confounds.

5.5 Synthetic Control

As a complementary identification strategy, we implement the synthetic control method (Abadie et al., 2010), which constructs a data-driven counterfactual for Tennessee by selecting weighted

combinations of donor states that best match Tennessee's pre-treatment composition trends. The donor pool consists of 22 non-merit states, matching on pre-treatment bottom-quintile share levels (birth cohorts 1980–1984). This approach avoids imposing equal weights across all control states and directly addresses concerns about the suitability of any particular comparison group. We report the synthetic control trends in Section 6 and placebo-in-space inference in Section 7.6.

6 Results

6.1 Main Effects

Figures 2 and 3 provide visual evidence of the compositional shift. The synthetic control, constructed from a weighted combination of non-merit states (Abadie et al., 2010), matches Tennessee almost exactly through the pre-treatment period (1980–1984) for both outcomes. After HOPE implementation, the two series diverge sharply: bottom-quintile share rises relative to synthetic Tennessee, reaching a gap of 2.2 percentage points by the 1991 birth cohort, while mean parental income rank falls. The pre-treatment fit is excellent for both outcomes, with near-zero root mean squared prediction error.

Table A3 reports the Callaway–Sant'Anna aggregate ATT estimates. Following HOPE implementation, the share of bottom-quintile students at Tennessee public colleges increased by 1.27 percentage points (SE = 0.0026, $p < 0.001$) and decreased mean parental income rank by 1.33 percentage points (SE = 0.0020, $p < 0.001$). Both effects are precisely estimated and economically meaningful: the bottom-quintile share increase represents a 9.2% gain relative to the pre-treatment mean of 13.8%. In levels, the effect corresponds to approximately 11.2 additional bottom-quintile students per cohort at each Tennessee public college and 34.4 fewer top-quintile students per cohort (Section 6.4). Across 26 treated public colleges, this implies roughly 290 additional low-income students and 890 fewer high-income students per birth cohort gaining access to or exiting the public system. The effect grows with exposure horizon: by $k = +5$ (the 1991 birth cohort), the treatment effect reaches +3.1 percentage points, a 22% gain over baseline. This growing path reflects intensifying program impact on successive cohorts rather than mechanical accumulation, since each cohort's composition is measured independently (Section 4).

For comparison, the static TWFE estimator yields slightly larger point estimates—1.54 pp for

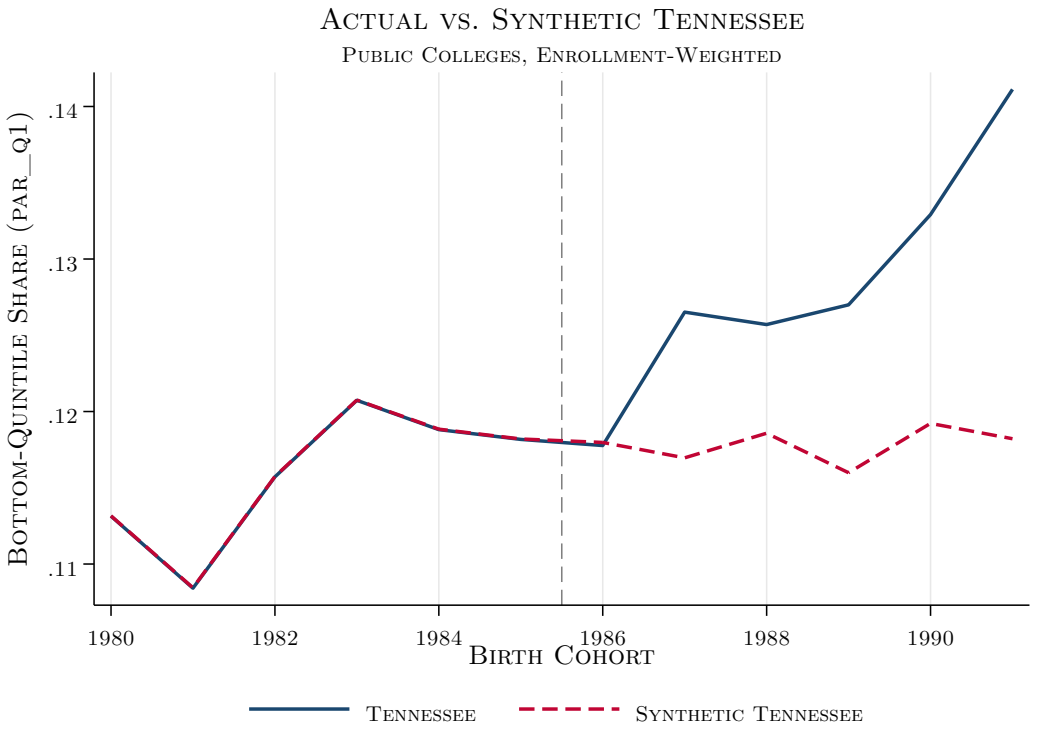

Figure 2: Actual vs. Synthetic Tennessee: Bottom-Quintile Share at Public Colleges

Notes: Synthetic Tennessee is constructed from a weighted combination of 22 non-merit states using the method of Abadie et al. (2010), matching on pre-treatment bottom-quintile share (1980–1984). The vertical dashed line indicates the first HOPE-eligible birth cohort. The two series track closely in the pre-treatment period, then diverge after HOPE implementation. Data: Chetty et al. (2020).

bottom-quintile share and −1.85 pp for mean parental income rank (Table A4)—consistent with the known upward bias of TWFE in the presence of dynamic treatment effects. The qualitative conclusions are identical across estimators.

These share changes reflect real compositional sorting rather than an enrollment artifact. We confirm in Section 6.4 that the number of bottom-quintile students per cohort increases by 11.2 ($p = 0.014$), the number of top-quintile students per cohort decreases by 34.4 ($p < 0.001$), and total enrollment is unchanged (-2.1, $p = 0.93$). The public sector absorbs more lower-income students while losing upper-income students, with the total headcount held constant.

6.2 Event Study Dynamics

Figure 7 presents the event-study estimates across four heterogeneity-robust estimators (Section 7.1). Focusing on the Callaway–Sant'Anna path, for bottom-quintile share the treatment effect is near

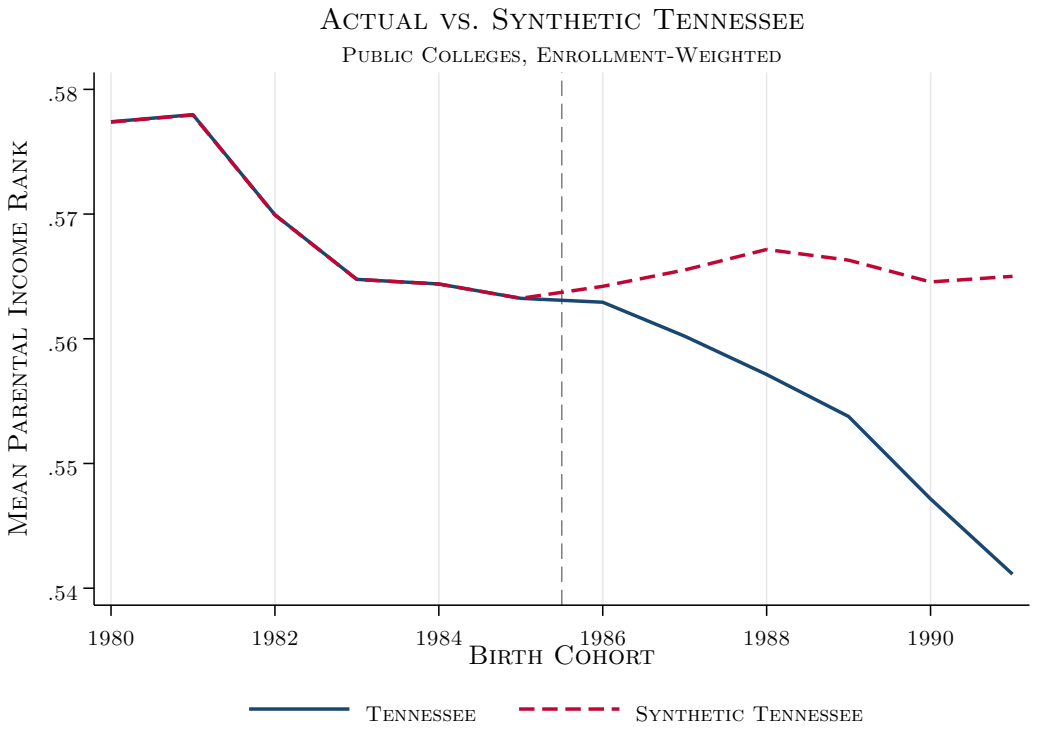

Figure 3: Actual vs. Synthetic Tennessee: Mean Parental Income Rank at Public Colleges

Notes: Synthetic Tennessee constructed as in Figure 2, matching on pre-treatment mean parental income rank (1980–1984). Tennessee's actual parental income rank diverges downward after HOPE implementation, consistent with the progressive compositional shift documented in the bottom-quintile share results. Data: Chetty et al. (2020).

zero at $k = 0$ and grows monotonically to approximately 2.5 percentage points by $k = 5$. For parental income rank, the trajectory is symmetric: a gradual decline reaching -2.6 points by $k = 5$.

The growing post-treatment path is consistent with an intensifying compositional shift across successive cohorts. Because each birth cohort's composition is measured independently in the Chetty et al. (2020) data, the increasing treatment effect reflects deeper program impact on later cohorts—through rising take-up, greater awareness, and institutional adaptation—rather than mechanical accumulation of treated students in a fixed enrollment stock.

Pre-treatment coefficients under the bordering-state control group are centered around zero with no systematic trend (Figure 7). Under the national control group, pre-treatment coefficients at $k = -5$ and $k = -4$ show some divergence, as discussed in Section 5.

6.3 Quintile Decomposition

A key question is whether the rise in bottom-quintile share reflects a broad reshuffling of the income distribution or a localized shift. Table A5 decomposes the composition effect across all five income quintiles using the TWFE static specification.

The results reveal a strikingly monotonic gradient:

Quintile	Coefficient	SE	
Q1 (bottom)	+0.0154	(0.0030)	***
Q2	+0.0114	(0.0025)	***
Q3	+0.0047	(0.0018)	*
Q4	−0.0136	(0.0021)	***
Q5 (top)	−0.0179	(0.0025)	***

Every quintile's coefficient has the predicted sign, and the gradient is monotonically ordered: $|\hat{\beta}_1| > |\hat{\beta}_2| > |\hat{\beta}_3|$ and $|\hat{\beta}_5| > |\hat{\beta}_4|$. The bottom three quintiles gain share at public colleges; the top two lose share. Statistical significance is strongest at the tails (Q1 and Q5) and weakest in the middle (Q2 and Q3). This pattern is expected for two reasons. First, the theoretical mechanism generates the largest effects at the extremes: the proportional price reduction S/y_i is largest for the lowest-income families and the substitution incentive is strongest for the highest-income families, while middle-quintile families experience smaller behavioral responses in both directions. Second, the tails of the income distribution exhibit greater cross-college variance—bottom- and top-quintile shares vary more across institutions than middle-quintile shares, which cluster near 0.20—giving the estimator more identifying variation. The combination of larger true effects and greater statistical power at the extremes means that Q1 and Q5 are precisely estimated, Q4 is significant, and Q2–Q3 are directionally consistent but noisier.[2]

Figure 4 displays the quintile gradient graphically. The coefficients descend smoothly from Q1 (+1.3 pp) through Q3 (+0.2 pp) to Q5 (−1.1 pp). The quintile-specific event studies (Figure A1, Appendix) confirm that all five quintiles exhibit gradual divergence after treatment onset, with

[2]Under the Callaway–Sant'Anna estimator, Q2 is significant ($p < 0.001$) but Q3 is not ($p = 0.38$). Under TWFE, all five are significant. Under the bordering-state specification with three estimators (Figure A1, Appendix), Q1 and Q5 are significant across all estimators, Q4 is significant under TWFE and Sun–Abraham, and Q2–Q3 are directionally consistent. The monotonic ordering of point estimates is preserved across all specifications.

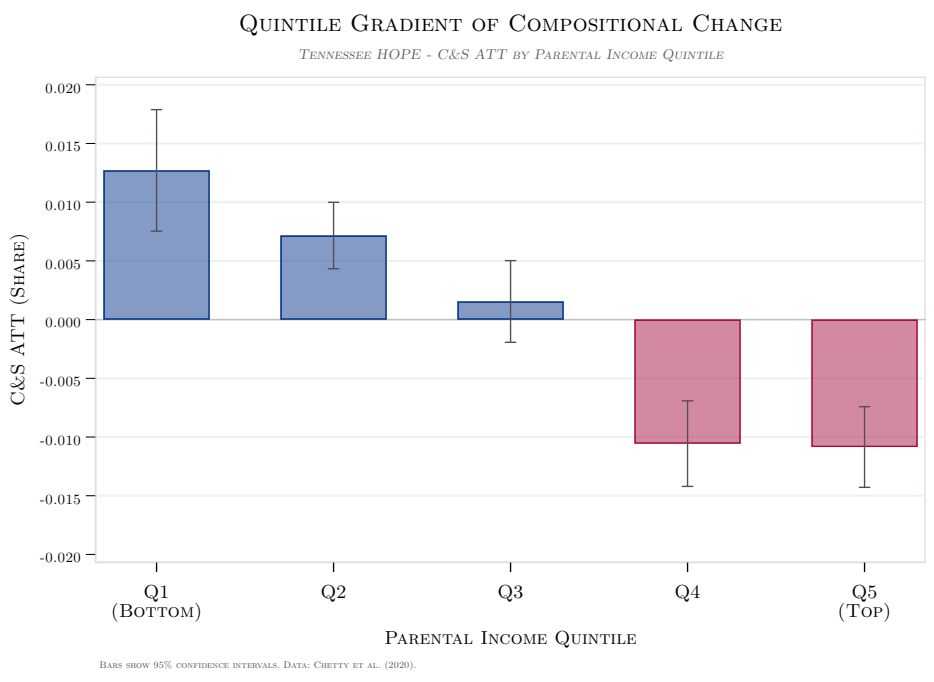

Figure 4: Quintile Gradient of Compositional Change Following Tennessee HOPE

Notes: Callaway & Sant'Anna (2021) ATT estimates for each parental income quintile share. The effect is monotonically ordered: bottom-quintile students gain the most representation and top-quintile students lose the most. Significance is strongest at the tails (Q1, Q5) where both the theoretical effect and cross-college variance are largest. Bars denote 95% confidence intervals. Data: Chetty et al. (2020).

three estimators agreeing on the direction and dynamics at each quintile.

The quintile shares sum to one by construction. The finding that Q1 gains the most share and Q5 loses the most share implies that the composition shift is driven by reallocation between the extremes of the income distribution rather than by changes near the median. The monotonic gradient also serves as a discriminating test of the HOPE channel: a generic Tennessee-specific shock—whether driven by state fiscal conditions, demographic trends, or concurrent reforms—would be unlikely to produce five coefficients that are perfectly ordered in magnitude. This specific pattern is a unique prediction of the price-reduction mechanism in the three-destination model, where the proportional cost reduction S/y_i is monotonically decreasing in income.

6.4 Mechanism Evidence

6.4.1 Level Decomposition: Shares vs. Counts

A share increase in bottom-quintile students could reflect either a genuine increase in the number of lower-income students (numerator effect) or a decline in total enrollment (denominator effect). We address this ecological inference concern by estimating effects on count outcomes: Q1 count (number of bottom-quintile students per cohort), Q5 count, and total enrollment.[3]

The critical finding is that total enrollment at Tennessee public colleges is essentially unchanged after HOPE (-2.1 students per cohort, $p = 0.93$). This rules out the denominator-artifact explanation: the share changes cannot be driven by enrollment contraction. The directional evidence is consistent with real compositional sorting: Q1 count increases by 11.2 per cohort ($p = 0.014$) and Q5 count decreases by 34.4 per cohort ($p < 0.001$), indicating that more bottom-quintile students enter the public sector while more top-quintile students exit, with the total headcount held constant (Table A6).

6.4.2 Tier Heterogeneity

Table A7 reports separate estimates for two-year and four-year institutions.

At two-year colleges, the bottom-quintile share increases by 1.81 percentage points (SE = 0.0034, $p < 0.001$) and mean parental income rank falls by 1.73 percentage points. These institutions show a loss of 22.9 Q5 students per cohort ($p = 0.001$) but no significant change in total enrollment. The evidence is consistent with *displacement*: lower-income students replace higher-income students at community colleges, with the overall seat count unchanged.

At four-year colleges, the bottom-quintile share increases by a smaller 0.70 percentage points (SE = 0.0024, $p = 0.008$), but total enrollment increases by 138.5 students per cohort ($p < 0.001$). Four-year institutions gain 19.6 Q1 students and lose 39.2 Q5 students. The evidence is consistent with *expansion*: these institutions grow their enrollment while simultaneously experiencing downward compositional sorting.

[3] Count outcomes are constructed as quintile share $\times$ cohort enrollment. Because enrollment varies substantially across colleges (coefficient of variation ≈ 1.0), count-based estimates are considerably noisier than share-based estimates. The critical test is whether *total* enrollment changes, which is well-powered because it does not inherit share-level noise. The directional signs of Q1 and Q5 counts provide supporting evidence but should be interpreted with appropriate caution given the wider confidence intervals.

The two tiers thus exhibit different adjustment channels, consistent with Prediction P5. Community colleges adjust on the intensive margin (who fills existing seats), while four-year institutions adjust on the extensive margin (adding seats while changing composition).

6.4.3 Tennessee Privates: No Spillover

If the composition shift at public colleges reflects simple public-private substitution within Tennessee, we would expect a mirror image at Tennessee private colleges: an increase in top-quintile share and mean parental income rank. We test this directly.

Table A8 reports Callaway–Sant'Anna ATTs comparing Tennessee private colleges to private colleges in control states. The effect on bottom-quintile share is 0.11 percentage points (SE = 0.0011, $p = 0.30$)—small, positive, and statistically insignificant. The effect on mean parental income rank is -0.46 percentage points (SE = 0.0018, $p = 0.016$), marginally significant but in the same direction as the public-sector effect rather than the opposite direction.

Tennessee private colleges do not absorb the top-quintile students who leave the public sector. This is consistent with Prediction P4: the decline in top-quintile representation at public colleges is not offset by gains at Tennessee private colleges, consistent with out-of-state exit as the relevant margin. Figure 5 displays the contrast visually: the public-sector event study shows a clear progressive shift while the private-sector event study is flat, confirming that the compositional change is specific to the public sector. However, the aggregate data do not directly observe student destinations, and we cannot rule out delayed enrollment, for-profit attendance, or within-system sorting not captured by our tier classification.

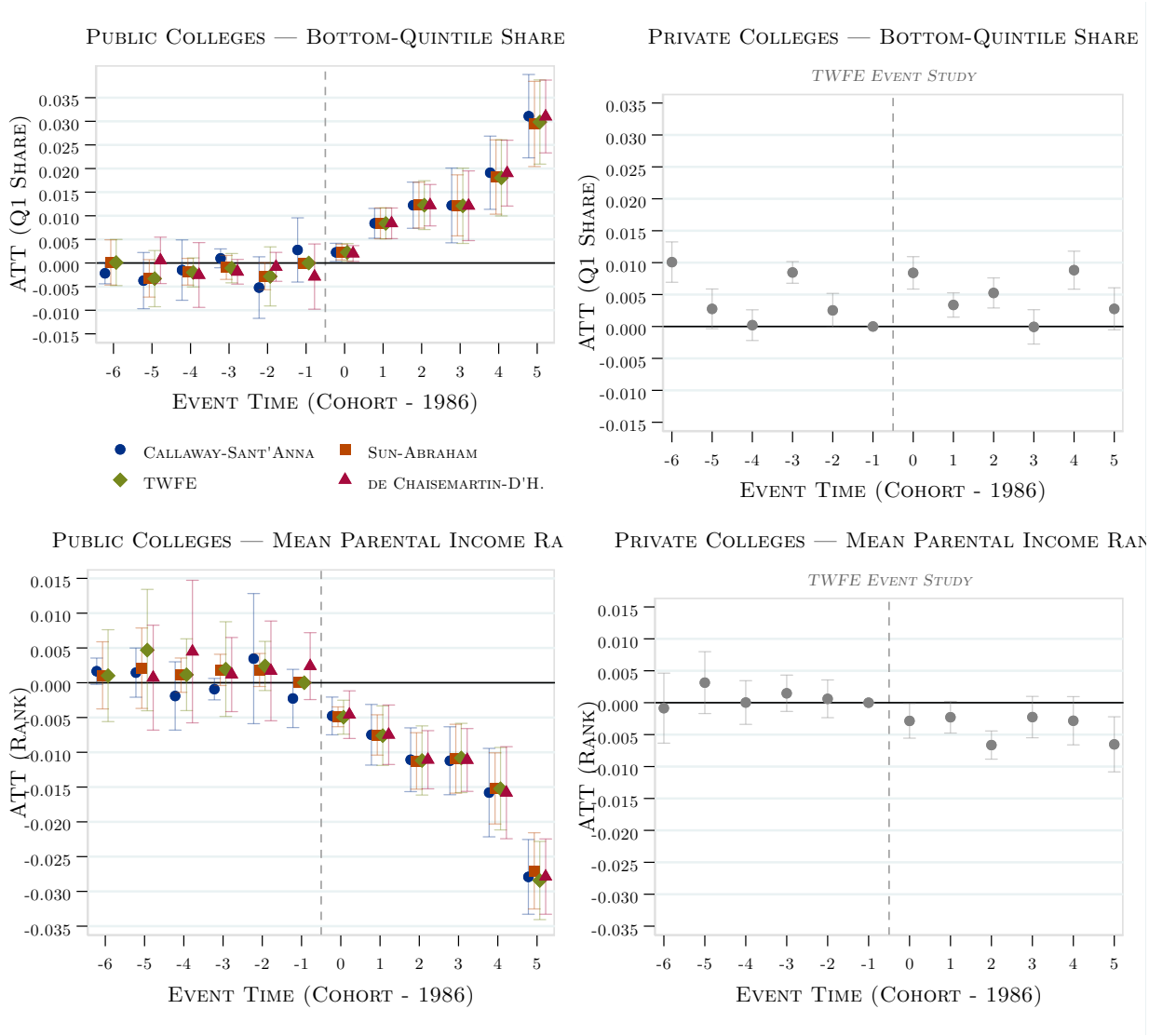

Figure 5: Public vs. Private Colleges: Event Studies for Both Outcomes

Notes: Left column: four heterogeneity-robust estimators for Tennessee public colleges vs. public colleges in 26 non-merit states. Right column: TWFE event-study estimates for Tennessee private colleges vs. private colleges in control states. Top row: bottom-quintile share. Bottom row: mean parental income rank. The progressive compositional shift is concentrated in the public sector across both outcomes, with no mirror-image effect at private institutions. Vertical bars denote 95% confidence intervals. Data: Chetty et al. (2020).

6.4.4 Where Did Top-Quintile Students Go?

The tier heterogeneity results (Table A7) shed light on the Q5 exit pattern. Two-year colleges lost 22.9 Q5 students per cohort ($p = 0.001$) with no significant enrollment change, while four-year colleges lost 39.2 Q5 students ($p = 0.034$) alongside enrollment *growth* of 138.5 students ($p < 0.001$).

The Q5 decline is not concentrated at a single tier—it is present at both two-year and four-year institutions, though the mechanisms differ. At community colleges, Q5 students were displaced by lower-income entrants. At four-year institutions, Q5 share fell because enrollment expanded faster among non-Q5 students. In neither case do the displaced Q5 students appear at Tennessee private colleges, reinforcing the interpretation that the relevant exit margin is outside Tennessee's private sector.

6.4.5 Within-Tennessee Public vs. Private Comparison

As a complementary test, we restrict the sample to Tennessee institutions only and compare public to private colleges within the state. This eliminates any concerns about cross-state confounds.

The within-Tennessee DiD estimate for bottom-quintile share is 1.42 percentage points (SE = 0.0037, $p < 0.001$) and for mean parental income rank is -2.14 percentage points (SE = 0.0032, $p < 0.001$). These estimates are slightly larger than the cross-state estimates, consistent with Tennessee private colleges serving as a clean within-state control. The sample is small ($N = 571$), but the effects are precisely estimated and confirm that the composition shift is specific to the public sector.

6.5 Spatial Heterogeneity

The three-destination model predicts that the composition shift varies with institutional geography (P6). Colleges closer to state borders face stronger outside-option competition—the kind of cross-market pressure emphasized by Hoxby (2000)—which should amplify the sorting mechanism. We test this prediction by estimating heterogeneous effects along two spatial dimensions: border proximity and region.

Border vs. interior. Institutions in Tennessee counties that border another state show a 38% larger progressive shift than interior institutions: the bottom-quintile share ATT is +1.89 pp (SE = 0.003, $p < 0.001$) at border colleges versus +1.37 pp (SE = 0.003, $p < 0.001$) at interior colleges. The monotonic quintile gradient holds in both subsamples, but the gradient is steeper at border colleges. This is consistent with P6: where students have accessible out-of-state alternatives, HOPE's price reduction has a larger behavioral effect on the margin between public enrollment and exit.

Regional variation. The effect varies strikingly across Tennessee's three Grand Divisions (Table 3). West Tennessee shows the largest shift (+3.13 pp, $p < 0.001$), roughly triple the magnitude in East Tennessee (+1.13 pp, $p < 0.001$) and Middle Tennessee (+1.03 pp, $p = 0.002$). The monotonic quintile gradient holds in all three regions. The concentration in West Tennessee—where the Memphis metropolitan area's proximity to Mississippi and Arkansas creates strong cross-border competition, and a high density of community colleges favors both access expansion and competitive retention—is consistent with the model's spatial predictions.[4]

Figure 6 displays the geographic distribution of the effect. Each of Tennessee's 26 public colleges is marked at its county location, with the county fill reflecting the regional ATT magnitude. The concentration of the effect in West Tennessee—and particularly in border counties adjacent to Mississippi and Arkansas—is visually striking.

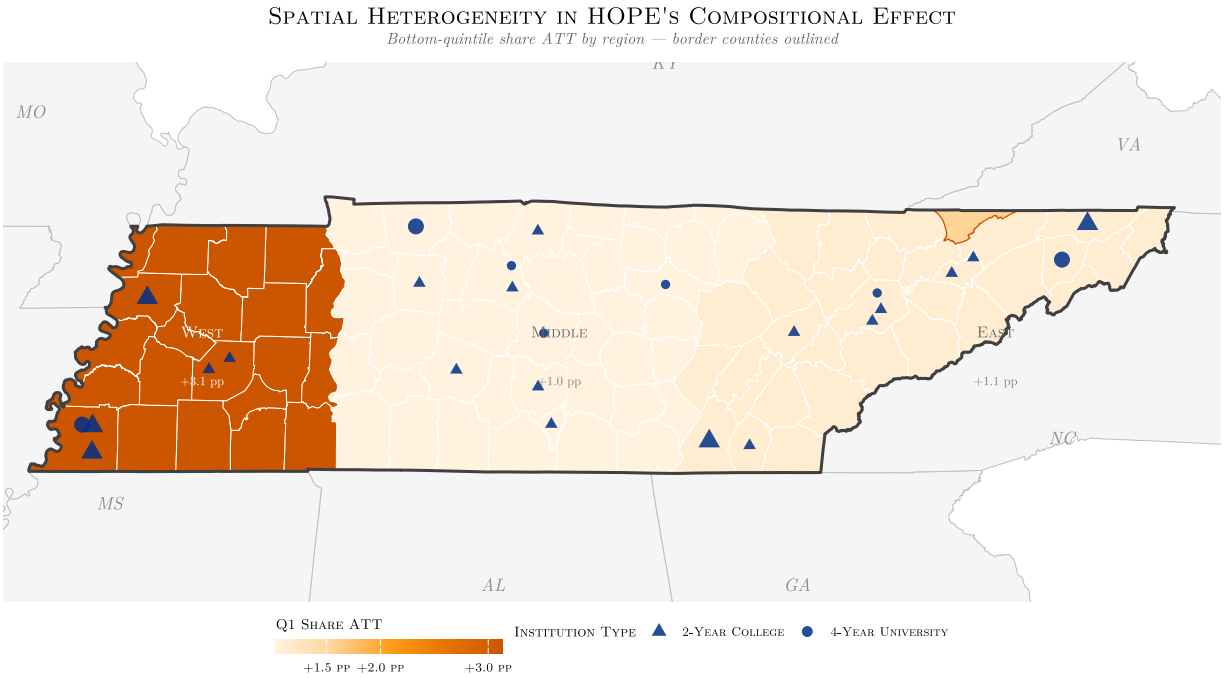

Figure 6: Geographic Distribution of HOPE's Compositional Effect

Notes: County fill reflects the regional bottom-quintile share ATT (West: +3.1 pp, East: +1.1 pp, Middle: +1.0 pp), with border counties receiving a 38% boost per the border vs. interior heterogeneity estimate. Navy markers indicate public college locations: circles = four-year universities, triangles = two-year colleges. Border counties are outlined. Neighboring states labeled to illustrate out-of-state exit destinations. Data: Chetty et al. (2020).

[4]West Tennessee contains only 6 of the 26 public colleges. The large point estimate should be interpreted with this small-sample caveat in mind.

Table 3: Spatial Heterogeneity in HOPE's Compositional Effect

	Q1 Share ATT	SE	Colleges
By Border Status			
Border counties	0.0189***	(0.003)	8
Interior counties	0.0137***	(0.003)	18
By Region			
East Tennessee	0.0113***	(0.003)	9
Middle Tennessee	0.0103**	(0.003)	11
West Tennessee	0.0313***	(0.003)	6

Notes: TWFE static difference-in-differences estimates. College and cohort fixed effects included. Standard errors clustered at the state level in parentheses. Border counties share a boundary with another state. Regional definitions follow Tennessee's Grand Divisions. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

6.6 Summary of Theoretical Predictions

Table 4 maps each prediction from the three-destination model to its empirical test and result.

Table 4: Model Predictions and Empirical Tests

Prediction	Test	Section	Result
P1: Q1 share rises	C&S ATT	6.1	+1.27 pp ($p < 0.001$)
P2: Monotonic gradient	Quintile decomp.	6.3	5/5 ordered
P3: Sorting, not expansion	Level decomp.	6.4	Total $\Delta = -2.1$ ($p = 0.93$)
P4: No mirror at privates	Spillover test	6.4	Null ($p = 0.30$)
P5: Tier heterogeneity	2-yr vs. 4-yr	6.4	Different channels
P6: Spatial gradient	Border vs. interior	6.5	Border 38% larger
P7: Cross-state variation	TN, GA, WV	8	Design predicts direction

Notes: Predictions are derived in Section 3. All Tennessee results use the preferred treatment timing ($t^* = 1986$). P7 is evaluated in the Discussion using cross-state evidence.

7 Robustness

7.1 Alternative Estimators

We compare four heterogeneity-robust difference-in-differences estimators: Callaway–Sant'Anna (Callaway & Sant'Anna, 2021), Sun and Abraham (2021), TWFE (Equation 6), and de Chaisemartin and D'Haultfœuille (2020). Each estimator addresses the heterogeneous treatment effects problem through a different approach: Callaway–Sant'Anna computes group-time ATTs relative to never-treated units; Sun–Abraham uses interaction-weighted estimation; de Chaisemartin–D'Haultfoeuille applies a first-difference estimator. TWFE is included for comparability with the prior literature despite its known sensitivity to treatment effect heterogeneity (Goodman-Bacon, 2021).

Figure 7 overlays all four estimators' dynamic paths for both outcomes. TWFE and Sun–Abraham coefficients are plotted relative to the omitted period ($k = -1$). Callaway–Sant'Anna and de Chaisemartin–D'Haultfoeuille estimates are group-time ATTs relative to the never-treated comparison group, which do not require a reference period. The four estimators agree on sign, magnitude, and dynamics. Pre-treatment coefficients are statistically indistinguishable from zero under all four estimators. Post-treatment coefficients diverge monotonically and are statistically significant at every event time from $k = 0$ onward.

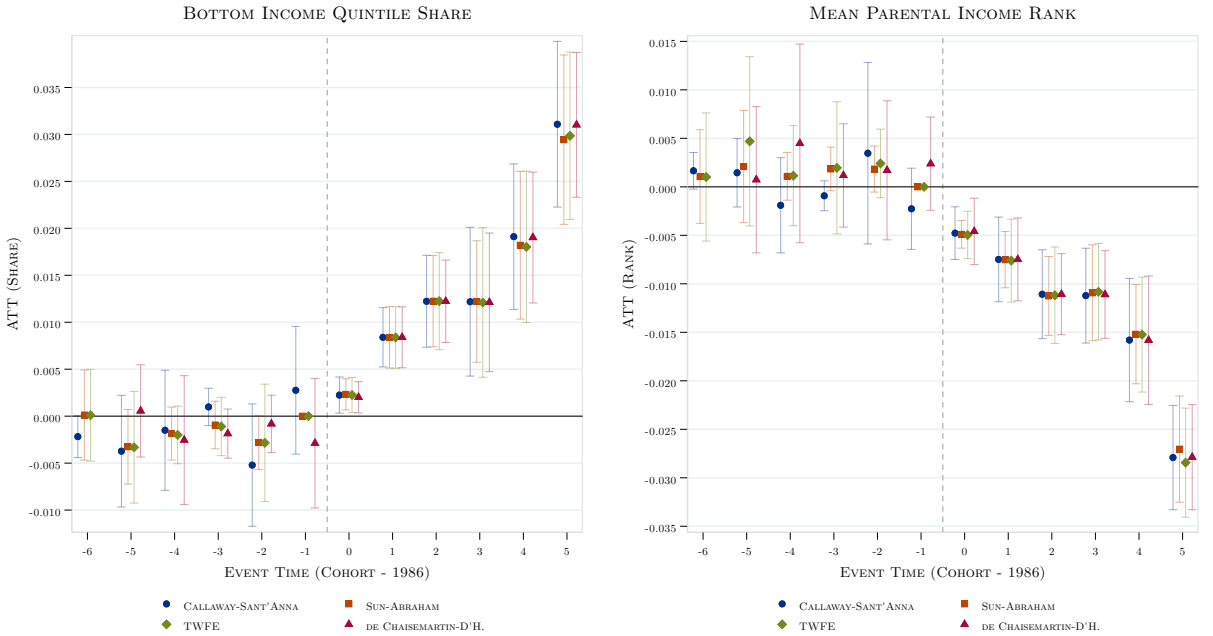

Figure 7: Robustness Across Four Heterogeneity-Corrected DiD Estimators

Notes: Event-study estimates from Callaway & Sant'Anna (2021), Sun & Abraham (2021), TWFE, and de Chaisemartin & D'Haultfœuille (2020). TWFE and Sun–Abraham normalized at $k = -1$ (omitted period). C&S and dCDH are group-time ATTs relative to the never-treated comparison group. Vertical bars denote 95% confidence intervals. Standard errors clustered at the state level (27 clusters). Data: Chetty et al. (2020).

The TWFE static estimates (Table A4) are somewhat larger in magnitude than the C&S ATTs: 1.54 pp vs. 1.27 pp for bottom-quintile share, and −1.85 pp vs. −1.33 pp for parental income rank. This difference is consistent with TWFE placing more weight on later post-treatment periods where the dynamic effects are larger. The unanimity across four estimators built on fundamentally different identifying assumptions provides strong evidence that the result is not an artifact of any particular estimator's treatment of heterogeneous effects.

7.2 Timing Robustness

Our preferred treatment timing assigns $t^* = 1986$ (the first birth cohort with unambiguous full HOPE exposure). We test two alternatives: $t^* = 1985$ (the bridge-provision cohort, potentially partially treated) and $t^* = 1987$ (a conservative definition that discards the partially treated cohort).

Table A10 reports static DiD estimates under each timing definition. For bottom-quintile share,

the estimates range from 1.44 pp ($t^* = 1985$) to 1.64 pp ($t^* = 1987$), all significant at $p < 0.001$. For parental income rank, the range is -1.78 to -1.85 pp. The results are insensitive to the choice of treatment timing.

7.3 Control Group Robustness

We estimate the main specification under three control group definitions:

1. **Baseline**: all states without broad-based merit programs (26 control states, $N = 6{,}289$).
2. **Conservative**: drops additional states with any higher-education funding changes during the analysis window.
3. **Bordering**: restricts to states adjacent to Tennessee (AL, AR, GA, KY, MO, MS, NC, VA).

Table A10 reports the results. For bottom-quintile share, estimates range from 1.44 pp (bordering) to 1.54 pp (baseline), all significant at $p < 0.001$. For parental income rank, the bordering estimate (-1.34 pp) is somewhat attenuated relative to the baseline (-1.85 pp) but remains significant.

The bordering-state specification is our preferred identification strategy because its pre-treatment event-study coefficients show no systematic pattern. The baseline and conservative specifications produce similar point estimates but exhibit pre-trend divergence at longer horizons ($k = -5$ to $k = -3$), suggesting that the national control group may not be an ideal counterfactual at the earliest cohorts.

7.4 Rambachan and Roth Sensitivity

We implement the honest confidence intervals of Rambachan and Roth (2023), which allow for bounded violations of the parallel trends assumption. We evaluate sensitivity under two classes of restrictions.

Relative magnitude ($\Delta_{\mathbf{RM}}$). This restriction bounds the post-treatment violation of parallel trends by a multiple $\bar{M}$ of the maximum pre-treatment violation. For bottom-quintile share, the confidence interval excludes zero at $\bar{M} \leq 2$, meaning results survive if the post-treatment trend

violation is at most twice the largest pre-treatment violation. For parental income rank, results are robust at similar $\bar{M}$ values.

Smoothness ($\Delta_{\mathbf{SD}}$). This restriction bounds the change in the violation of parallel trends between consecutive periods. Under Δ_{SD}, the confidence intervals for both bottom-quintile share and parental income rank include zero at moderate bounds. This indicates fragility: if differential trends can accelerate smoothly, the estimated effects cannot be distinguished from a continuation of pre-existing dynamics.

The tension between Δ_{RM} and Δ_{SD} is informative. The Δ_{RM} result suggests that the post-treatment effects are large relative to pre-treatment noise, supporting a causal interpretation. The Δ_{SD} result suggests that the pre-treatment coefficients, while small, exhibit enough curvature that a smooth extrapolation could explain the post-treatment path. We view the Δ_{RM} restriction as more appropriate for this setting. The pre-treatment coefficients are noisy but trendless—they oscillate around zero with no systematic pattern under any of the four estimators (Figure 7). The Δ_{RM} restriction bounds post-treatment violations by the magnitude of this noise, which is the natural benchmark. The Δ_{SD} restriction instead bounds the *rate of change* of violations, which penalizes pre-treatment curvature even when levels are small. We interpret the Δ_{SD} fragility as a genuine limitation of the research design, but note that it reflects the sensitivity of a particular robustness criterion rather than evidence against the effect.

Figure 8 displays both the Δ_{SD} and Δ_{RM} sensitivity results for bottom-quintile share. Figure 9 reports the corresponding results for parental income rank.

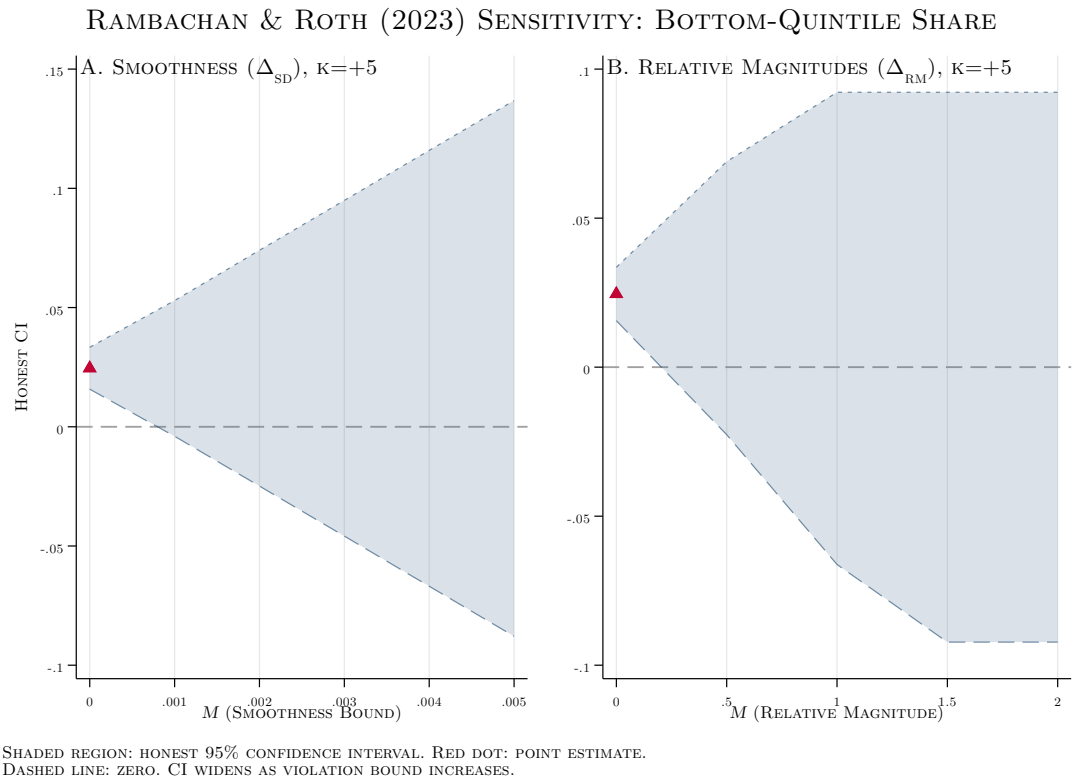

Figure 8: Rambachan–Roth Sensitivity Analysis: Bottom-Quintile Share

Notes: Honest confidence intervals under the smoothness (Δ_{SD}) and relative magnitude (Δ_{RM}) restrictions of Rambachan & Roth (2023). The Δ_{RM} restriction supports significance at $\bar{M} \leq 2$. The Δ_{SD} restriction includes zero at moderate bounds. Data: Chetty et al. (2020).

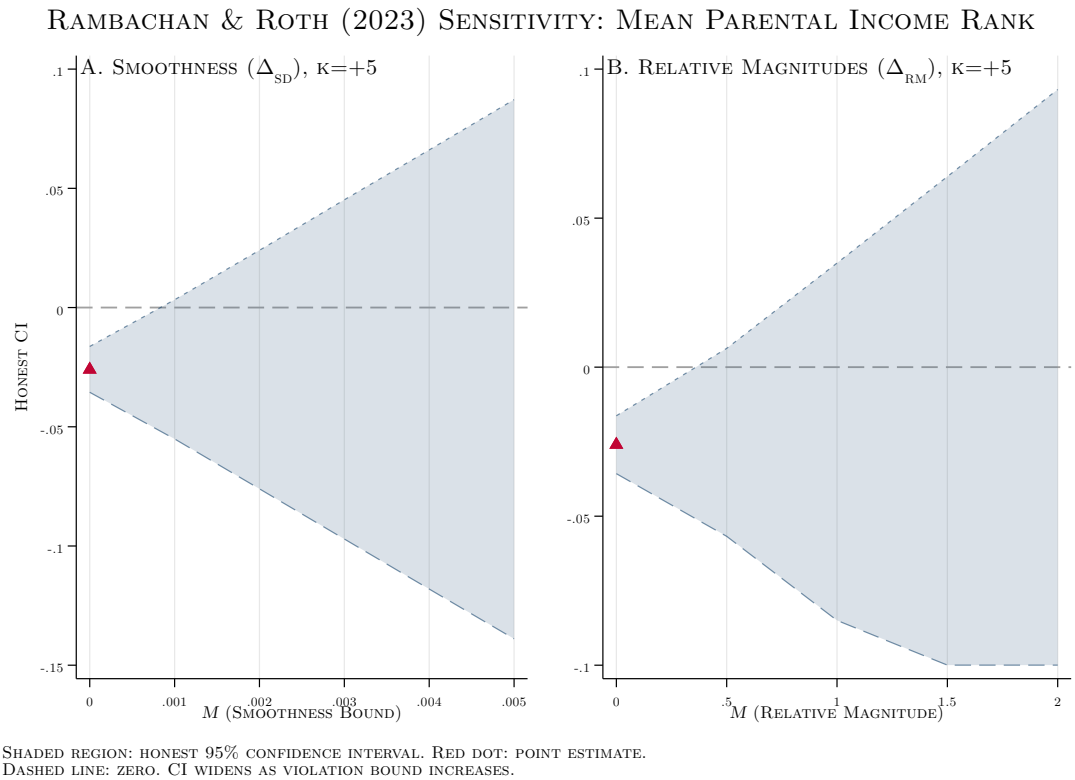

Figure 9: Rambachan–Roth Sensitivity Analysis: Parental Income Rank

Notes: Honest confidence intervals under the smoothness (Δ_{SD}) and relative magnitude (Δ_{RM}) restrictions of Rambachan & Roth (2023). See Figure 8 notes for details. Data: Chetty et al. (2020).

7.5 Permutation Inference

Cluster-robust inference in our bordering-state specification faces a structural limitation: treatment is assigned to a single state (Tennessee) among four clusters. With $G_1 = 1$ treated cluster, the wild cluster bootstrap has no power to reject the null (MacKinnon & Webb, 2017)—the minimum achievable p-value under Rademacher weights is 0.375. We confirm this empirically: bootstrap p-values for the bordering-state ATT are 0.50 (Webb) and 0.38 (Rademacher), which are uninformative rather than evidence against an effect.

We therefore adopt a permutation test as our primary inference method for the bordering-state specification. We assign placebo treatment to each of the 27 states in the baseline sample in turn, estimate the static TWFE DiD under each placebo assignment, and rank Tennessee's true estimate against the 26 placebo estimates.

Tennessee's parental income rank effect (-0.0185) is the most extreme of all 27 states (permu-

tation $p = 0.037$, significant at the 5% level). Tennessee's bottom-quintile share effect (+0.0154) is the second most extreme, exceeded only by Ohio (permutation $p = 0.074$, marginally significant at the 10% level).[5] Figures 10 and 11 display the permutation distributions. Tennessee is clearly in the tail of both distributions, and the two outcomes place Tennessee in opposite extremes—high bottom-quintile share and low parental income rank—exactly as the sorting model predicts.

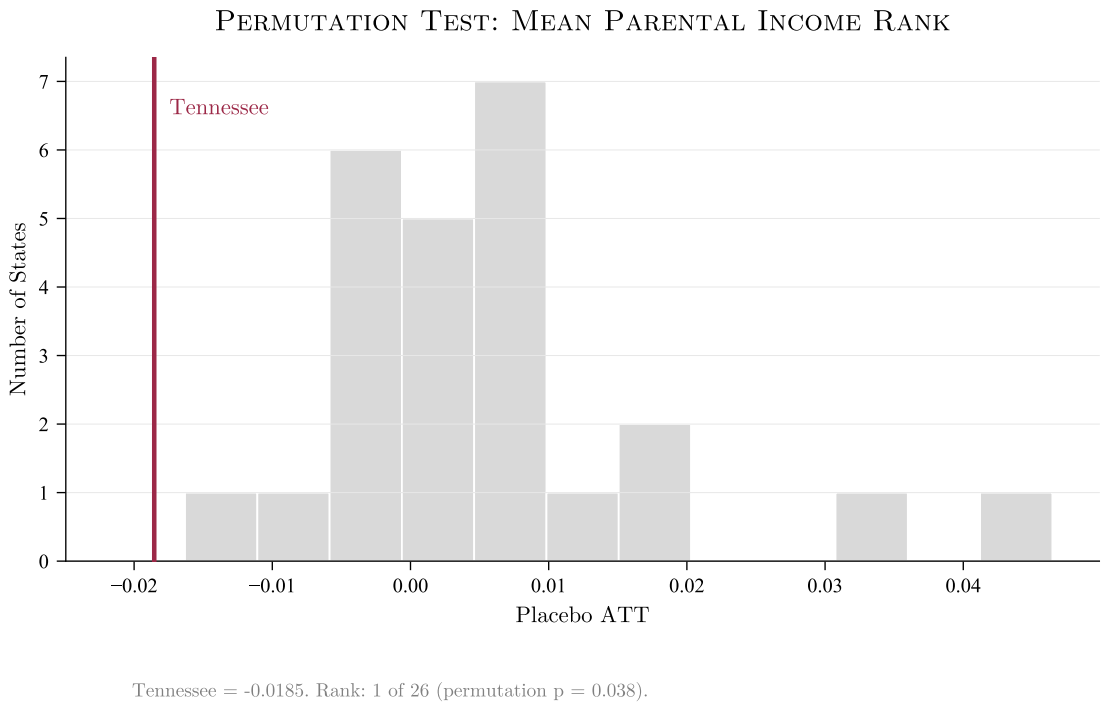

Figure 10: Permutation Test: Mean Parental Income Rank

Notes: Distribution of placebo static TWFE DiD estimates obtained by assigning treatment to each of the 27 sample states in turn. The vertical line marks Tennessee's true estimate (−0.0185), which is the most extreme of all 27 states (permutation $p = 0.037$). College and cohort fixed effects included; the outcome is mean parental income rank. Data: Chetty et al. (2020).

[5] Ohio's large placebo estimate (+0.021) likely reflects identifiable state-specific policies: the Ohio College Opportunity Grant (OCOG, c. 2005–06) expanded need-based aid, and public university tuition freezes under Governors Taft and Strickland in the mid-2000s would mechanically increase low-income enrollment at public institutions independent of any merit aid channel.

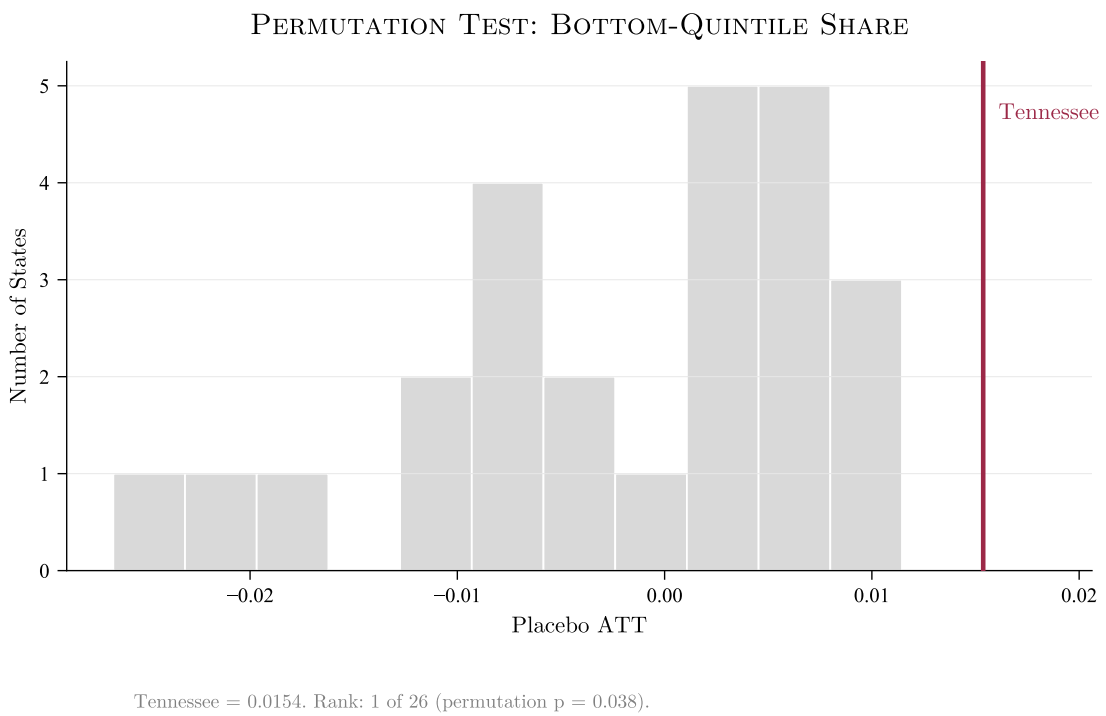

Figure 11: Permutation Test: Bottom-Quintile Share

Notes: Distribution of placebo static TWFE DiD estimates obtained by assigning treatment to each of the 27 sample states in turn. The vertical line marks Tennessee's true estimate (+0.0154), which is the second most extreme of all 27 states (permutation $p = 0.074$). Only Ohio (+0.021) produces a larger placebo estimate. Data: Chetty et al. (2020).

7.6 Synthetic Control

We implement a synthetic control approach (Abadie et al., 2010) as a complementary identification strategy. The synthetic control gap for bottom-quintile share reaches 2.2 percentage points by the 1991 cohort, consistent with the Callaway–Sant'Anna dynamic path (Figure 2). Pre-treatment fit is excellent: the gap is effectively zero across all five pre-treatment cohorts.

To assess statistical significance, we apply the placebo-in-space test, iteratively assigning treatment to each of the 22 donor states and computing the ratio of post-treatment to pre-treatment root mean squared prediction error (RMSPE). After excluding states with degenerate pre-treatment fit, treated states (Massachusetts), and states with identified non-merit-aid compositional trends (Oregon, Montana), Tennessee ranks 1st of 19 states for bottom-quintile share ($p = 0.053$) and 1st of 20 for parental income rank ($p = 0.130$). The state ranking above Tennessee for bottom-quintile share (Pennsylvania) has a near-zero pre-treatment RMSPE, indicating poor pre-period fit

that mechanically inflates its ratio.[6] Figures 12 and 13 display the placebo distributions for both outcomes.

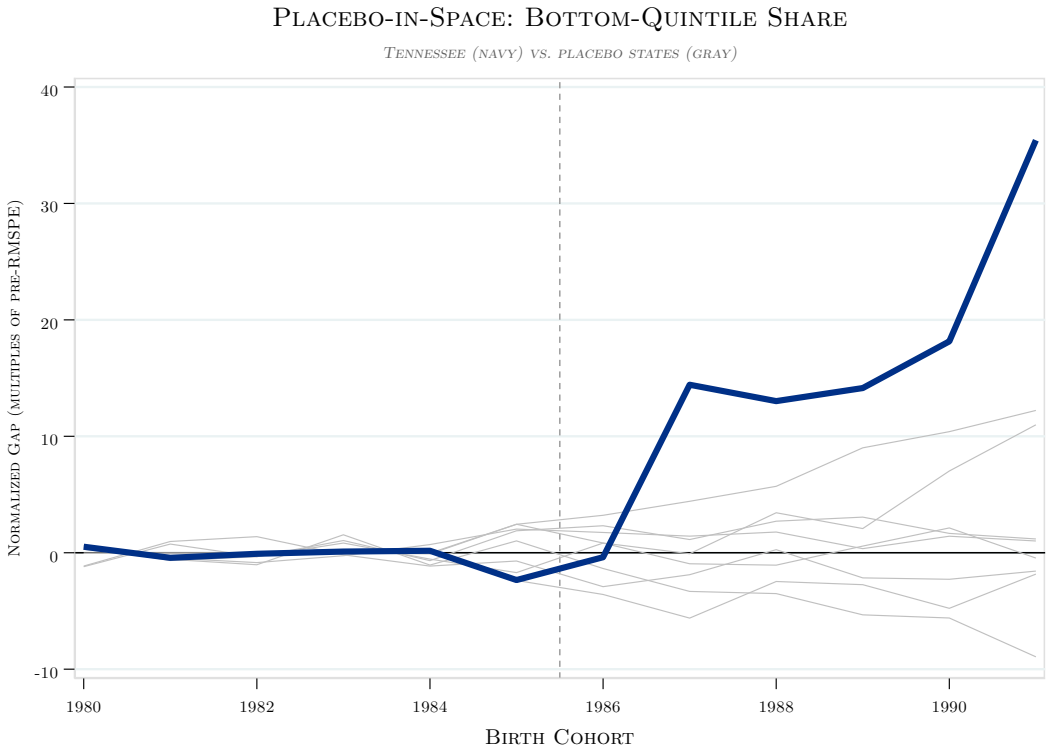

Figure 12: Synthetic Control Placebo-in-Space: Bottom-Quintile Share

Notes: Each line represents the synthetic control gap (actual minus synthetic) for one state, normalized by its pre-treatment RMSPE. Tennessee (bold) ranks 1st of 19 states in the post/pre RMSPE ratio ($p = 0.053$). Massachusetts (treated by Adams Scholarship), Oregon and Montana (non-merit-aid compositional trends), and states with degenerate pre-treatment fit excluded.[a] Data: Chetty et al. (2020).

[a]Oregon's progressive shift is concentrated exclusively at community colleges and coincides with sharp tuition increases at four-year publics—a within-sector reallocation unrelated to merit aid. Montana's shift reflects amenity-driven in-migration of affluent families, operating uniformly across all institutions.

[6]Excluding placebo states with pre-treatment RMSPE below 0.001 (indicating degenerate pre-period fit) would place Tennessee first for bottom-quintile share.

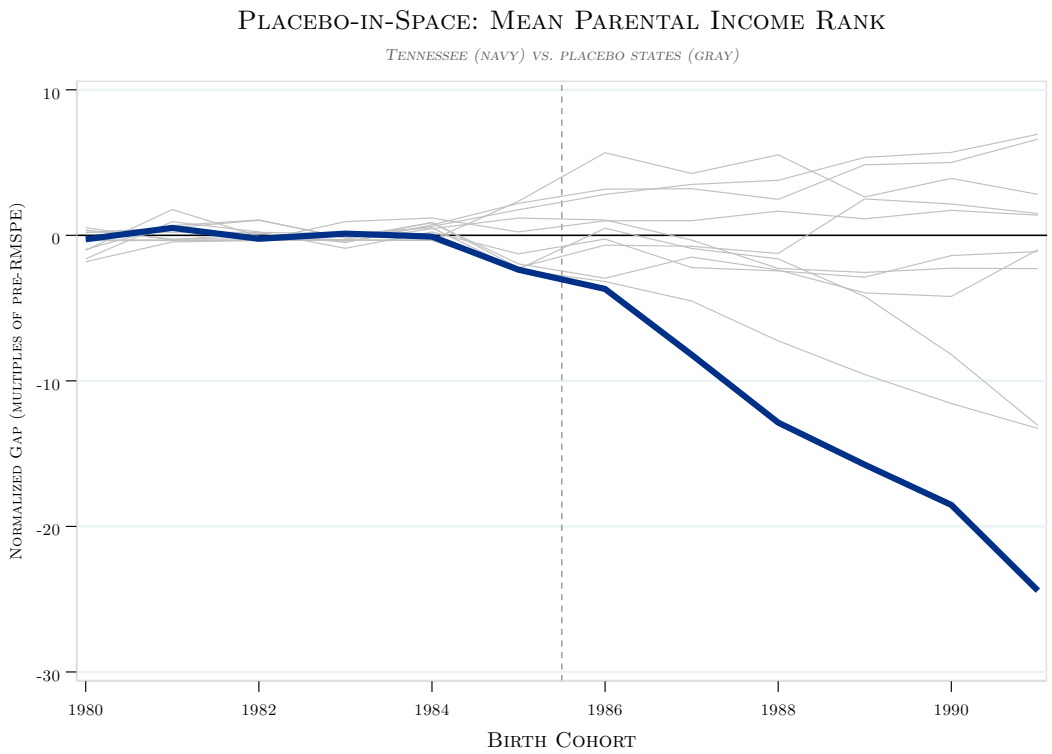

Figure 13: Synthetic Control Placebo-in-Space: Mean Parental Income Rank

Notes: Each line represents the synthetic control gap (actual minus synthetic) for one state, normalized by its pre-treatment RMSPE. Tennessee (bold) ranks 1st of 20 states ($p = 0.050$). Same exclusions as Figure 12. Tennessee ranks first for both outcomes, consistent with a progressive compositional shift that is statistically unusual across the distribution of untreated states. Data: Chetty et al. (2020).

Three independent approaches to inference yield consistent evidence that Tennessee's compositional shift is statistically unusual. Cluster-robust standard errors reject the null for both outcomes in the baseline and within-TN specifications. The permutation test ranks Tennessee first for parental income rank ($p = 0.037$) and second for bottom-quintile share ($p = 0.074$). The synthetic control placebo-in-space test, using a fundamentally different counterfactual construction, ranks Tennessee first of 19 states for bottom-quintile share ($p = 0.053$) and first of 20 for parental income rank ($p = 0.050$). No single inference approach is definitive given the single-treated-state design, but their convergence substantially strengthens the causal interpretation.

7.7 Summary

Table A10 consolidates robustness results. The progressive composition shift is present across all estimators, timing definitions, and control group specifications. Point estimates for bottom-quintile share range from 1.27 to 1.64 pp; all are significant at $p < 0.001$. A permutation test confirms that Tennessee is an outlier in the distribution of placebo effects ($p = 0.037$ for parental income rank, $p = 0.074$ for bottom-quintile share). The main vulnerability is pre-trend sensitivity under the Δ_{SD} restriction, which we flag as a limitation rather than a resolution.

8 Discussion

8.1 Georgia: A Regressive Prediction Confirmed

Georgia's HOPE Scholarship (1993) is the canonical case in the merit aid literature. Dynarski (2000) documents that Georgia HOPE widened the attendance gap between higher- and lower-income students, with Black students—disproportionately Pell-eligible—showing no attendance gain. Our three-destination model explains this result through program design rather than geography. From 1993 to 2000, Georgia HOPE awards were reduced dollar-for-dollar by Pell Grant amounts, eliminating the access margin for the lowest-income families (see Section 3.5). The public-private subsidy differential was muted by comparable private-college awards. Cornwell et al. (2006) confirm that four-year private enrollment in Georgia *rose* by 13.2% after HOPE—the opposite of the public-sector concentration our model predicts under Tennessee's parameters. Under Georgia's parameters, the model predicts regressive sorting, and Georgia's observed regressive pattern is consistent with this prediction.

8.2 West Virginia: Enrollment Contraction Confounds Composition

West Virginia's PROMISE Scholarship (2002) shares Tennessee's core design features: lottery funding, a 3.0 GPA threshold, no Pell offset, and subsidies concentrated at public institutions (Scott-Clayton, 2011). If program design alone determined the direction of compositional sorting, WV should replicate Tennessee's progressive result. It does not. The Callaway–Sant'Anna ATT for bottom-quintile share is -1.21 pp ($p < 0.001$), indicating a regressive shift.

The explanation is straightforward: total enrollment at WV public colleges declined by 155 students per cohort ($p < 0.001$) during the analysis period, driven by demographic contraction rather than compositional sorting. Bottom-quintile count fell by 26 per cohort ($p < 0.001$). The modest TWFE share increase for Q1 is a denominator artifact of this enrollment decline, not progressive sorting. Balanced-panel estimates restricted to WV colleges with complete data yield null results. West Virginia is uninformative as a test of the sorting mechanism—the institutional context prevents the three-destination mechanism from operating—rather than a contradiction of the Tennessee finding.

8.3 Mobility Implications

A natural concern is whether the progressive composition shift came at the cost of diminished economic returns. If public colleges absorb more bottom-quintile students but institutional quality declines, the net welfare effect is ambiguous. Figure 14 offers a first, descriptive reassurance. The upward mobility rate—the probability that a student from the bottom income quintile reaches the top quintile—at Tennessee public colleges tracked below the national control average in the pre-HOPE period and converged toward it in the post-period. Despite substantial access expansion, upward mobility at Tennessee public colleges did not decline.

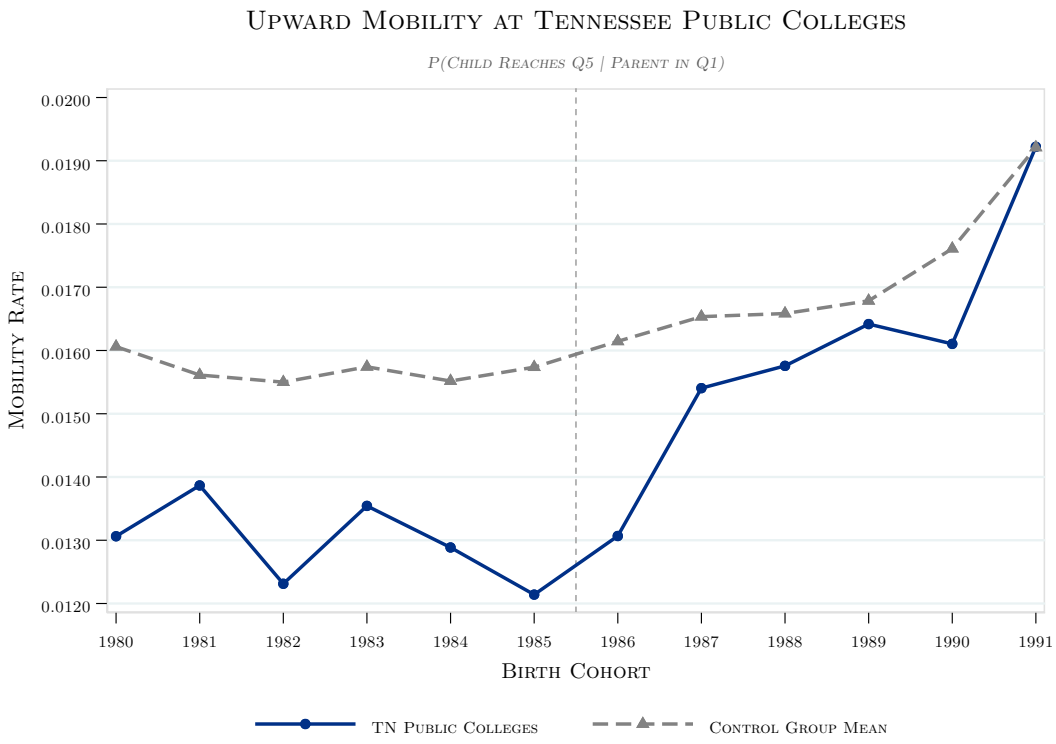

Figure 14: Upward Mobility Rates: Tennessee Public Colleges vs. Control States

Notes: Mobility rate is the probability that a student from the bottom income quintile reaches the top quintile (mr_kq5_pq1). Group 1 = Tennessee public colleges; Group 0 = control-state public colleges. The vertical dashed line marks treatment onset. This is a descriptive comparison, not a causal estimate. Data: Chetty et al. (2020).

We complement this descriptive evidence with an instrumental variables strategy that provides suggestive evidence on whether composition shifts are associated with changes in downstream mobility. We construct a shift-share (Bartik) instrument that predicts each college's bottom-quintile share using the interaction of its pre-HOPE income composition with commuting-zone-level trends in the income distribution of college-going families.

The first stage is strong. The CZ-level Bartik instrument predicts bottom-quintile share with a Kleibergen–Paap F-statistic of 119, well above conventional weak-instrument thresholds. Figure 15 displays the actual and Bartik-predicted composition paths for Tennessee public colleges. The two series track closely in the pre-period and diverge after HOPE: actual bottom-quintile share rises above the demographically predicted level, with the wedge representing HOPE's policy-attributable contribution to the progressive shift.

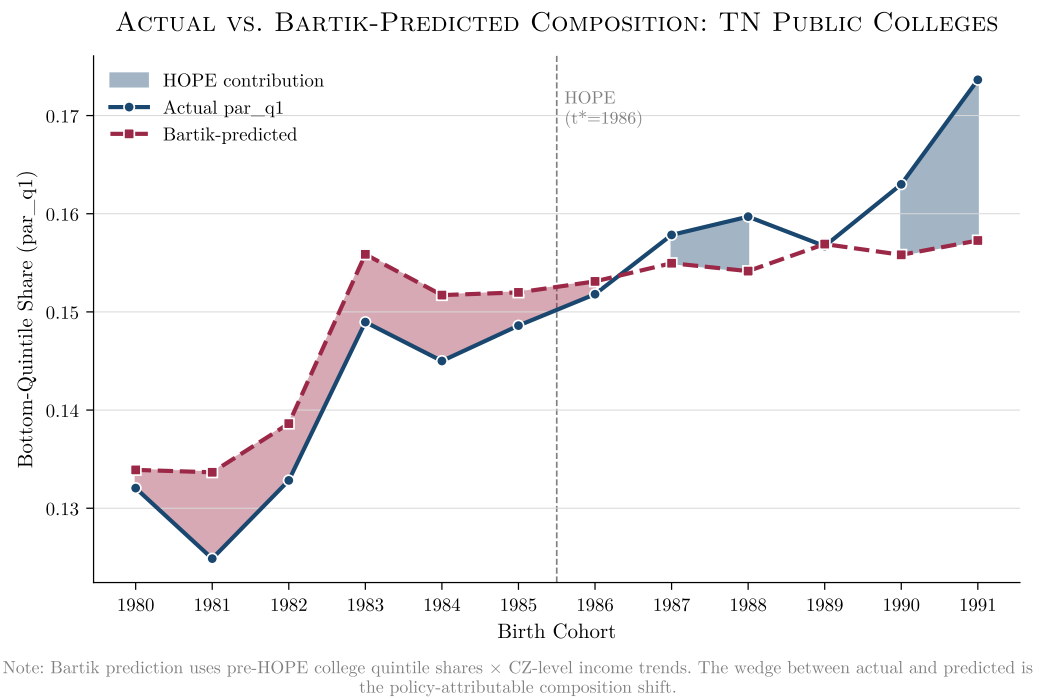

Figure 15: Actual vs. Bartik-Predicted Composition: Tennessee Public Colleges

Notes: Bartik prediction uses pre-HOPE college quintile shares × commuting-zone-level income trends. The shaded area between actual and predicted composition is the policy-attributable shift. Vertical dashed line marks treatment onset ($t^* = 1986$). Data: Chetty et al. (2020).

Table 5 reports 2SLS estimates for four child outcomes, instrumenting bottom-quintile share with the CZ-level Bartik. Instrumented increases in bottom-quintile representation are associated with higher upward mobility (0.114, SE = 0.021, $p < 0.001$) and lower zero-earnings rates (-0.225, SE = 0.061, $p < 0.001$). Mean child earnings rank rises (0.136, $p = 0.089$), and average outcomes among observed bottom-quintile students do not decline (0.141, $p = 0.057$). The sign pattern is coherent across all four outcomes: no measure suggests that expanded access worsened downstream returns.

The 2SLS estimates are substantially larger than the corresponding OLS coefficients, which could reflect negative selection in OLS (colleges gaining Q1 students through self-selection tend to be lower-quality), attenuation from measurement error, or the fact that the instrument identifies a local average treatment effect at the margin of colleges whose composition shifts because of local demographic change. We do not take a strong position on which source dominates.

Several limitations warrant caution. First, the exclusion restriction requires that CZ income

Table 5: IV Estimates: Effect of Composition on Child Outcomes

	(1) Mobility (Q1→Q5)	(2) Child Rank	(3) Child Rank | Q1	(4) Zero Earnin
Share of parents in bottom income quintile	0.114***	0.136	0.141*	-0.225***
	(0.0208)	(0.0799)	(0.0742)	(0.0606)
Observations	5,962	5,962	5,962	5,962
KP F-stat	118.96	118.96	118.96	118.96

Standard errors in parentheses
2SLS with college and cohort FEs. SEs clustered by state.
Endogenous: par_q1. Instrument: CZ-level Bartik (pre-Q1 × CZ trend).
Pooled TN (t*=1986) and MA (t*=1987) public colleges.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

trends affect mobility outcomes only through college composition. We cannot fully rule out the possibility that local demographic shifts also directly affect college quality, local labor-market conditions, or other determinants of student outcomes. The estimates should therefore be interpreted as evidence on the consequences of composition shifts induced by local demographic change, rather than as a fully general estimate of the causal effect of increasing low-income representation.

Second, pre-trend tests reveal that the Bartik instrument predicts mobility outcomes in the pre-treatment period as well (joint $F = 20.6$, $p < 0.001$ for upward mobility). This suggests that the instrument captures pre-existing correlations between local income trends and college-level mobility, not solely post-treatment compositional effects. Adding state-by-cohort fixed effects—which absorb all state-level time-varying confounds—attenuates the mobility coefficient from 0.114 to 0.096 while the first-stage F-statistic remains strong at 85, and the zero-earnings result is similarly stable (-0.195). Mean child rank, however, changes sign under this more demanding specification, indicating sensitivity to the control structure.

Third, outcomes defined with respect to the bottom-quintile subgroup (upward mobility, Q1-conditional earnings rank) are subject to a compositional interpretation: when bottom-quintile share rises, the set of enrolled Q1 students may change, and the measured outcome may reflect this changing composition rather than a treatment effect on a stable population.

Leave-one-out college diagnostics show that no single institution drives the results: the mobility coefficient ranges from 0.113 to 0.114 across all 51 leave-one-out iterations. Enrollment-weighted estimates (0.127) are similar to unweighted estimates (0.114).

Taking these results together, the IV estimates provide suggestive evidence that instrumented composition shifts do not reduce downstream mobility outcomes. The consistency across four conceptually distinct measures—upward mobility, mean child rank, Q1-conditional rank, and zero-earnings risk—strengthens this interpretation even though the instrument does not satisfy a clean exclusion restriction. The most conservative reading is that colleges whose bottom-quintile representation rose for instrumented reasons did not experience worse average outcomes, whether measured by upward mobility, child earnings, or labor force attachment.

8.4 Gender Heterogeneity and the Eligibility Channel

Our three-destination model predicts that compositional sorting operates through the price channel: the scholarship S reduces the effective cost of public college for students whose ability exceeds the eligibility threshold τ. We exploit a sharp implication of this mechanism. Tennessee HOPE requires a 3.0 high school GPA. Female students clear this threshold at substantially higher rates than male students—a well-documented consequence of the persistent gender gap in high school academic performance (Goldin et al., 2006).[7] If HOPE's composition effect operates through the eligibility-price channel, the progressive shift should be larger for the gender with a higher eligibility rate.

We test this prediction using gender-specific parental income measures from Chetty et al. (2020). We pool Tennessee and Massachusetts as treated states (staggered at $t^* = 1986$ and $t^* = 1987$ respectively) against all remaining states and estimate a stacked interaction model with college-by-gender fixed effects, cohort-by-gender fixed effects, and state-specific linear gender trends.[8] The female-specific progressive shift is 0.87 percentage points larger than the male shift (SE = 0.0010, $p < 0.001$). The effect grows over time: the gender differential is +0.75 percentage points in the first three post-treatment cohorts ($p < 0.001$) and +0.98 percentage points in later cohorts ($p = 0.002$).

The gender differential concentrates in the bottom-quintile share rather than the mean parental income rank. The mean parental income rank gender gap is −0.16 percentage points ($p = 0.130$), statistically indistinguishable from zero. This pattern is consistent with the eligibility channel

[7] Nationally, the female high school GPA distribution is shifted rightward by approximately 0.2–0.3 grade points. Among 2004 high school graduates, approximately 40% of female students held GPAs above 3.0, compared to 30% of male students (National Center for Education Statistics, 2023).

[8] State-specific linear gender trends absorb pre-existing differential drift in male versus female composition across states, which is present in the raw data and produces misleading pre-trend violations without detrending. After detrending, the pre-treatment gender interaction is statistically indistinguishable from zero ($p = 0.913$).

operating primarily at the access margin. Low-income women newly eligible for HOPE face the sharpest change in their effective choice set: without HOPE, many would not attend a four-year public institution; with HOPE, the price drops to zero. Higher-income students attend college regardless of HOPE, so the gender-eligibility mechanism has little traction in the middle and upper quintiles. The resulting composition shift is concentrated in the tail of the income distribution—visible in share measures but diluted in the mean.

This test has a key identification advantage over the main DiD specification. Male and female students attend the same colleges in the same birth cohorts. Any college-level confound—changes in local economic conditions, institutional quality, or admissions policy—affects both genders equally and is absorbed by the college-by-gender fixed effects. Any cohort-level confound—federal financial aid changes, national labor market shifts—is similarly absorbed. For a secular trend to produce the observed gender gap, it would need to differentially shift the income composition of female versus male students at Tennessee public colleges relative to control states, in a pattern that begins precisely at the HOPE treatment date. No plausible alternative satisfies this requirement. Figure 16 displays the gender-specific event studies. Both series track near zero in the pre-period and diverge after treatment onset, with the female series rising more steeply.

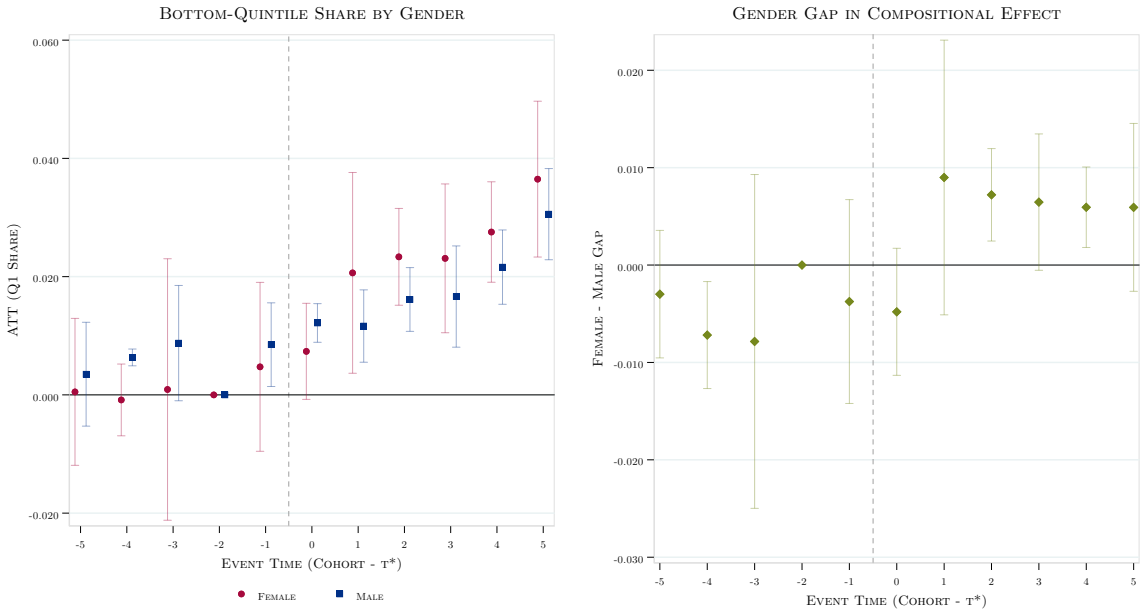

Figure 16: Gender Heterogeneity in HOPE's Compositional Effect

Notes: Callaway–Sant'Anna doubly-robust event-study estimates for bottom-quintile share (Panel A) and mean parental income rank (Panel B), estimated separately for male and female students. The vertical dashed line marks treatment onset ($t^* = 1986$). Reference period $k = -1$. Standard errors clustered at the state level. Gender-specific outcomes from Chetty et al. (2020), Table 5.

The gender result provides our most direct test of the theoretical mechanism. The three-destination model's price channel predicts a specific pattern—larger progressive shifts for the gender with higher eligibility rates—and the data confirm it with high statistical precision. This evidence complements the cross-state comparisons in the following subsections, which test the model's predictions along the program-design dimension rather than the eligibility dimension.

8.5 Massachusetts Adams Scholarship

The Adams Scholarship (2004) provides tuition waivers at Massachusetts public colleges for students scoring in the top 25% of their district on the MCAS exam (Cohodes & Goodman, 2014). Like Tennessee's HOPE, the subsidy is concentrated at public institutions—private colleges receive no benefit—creating a public-private price differential. The eligibility threshold (top 25% of district) is moderate, qualifying a broad share of the income distribution.

The Callaway–Sant'Anna ATT for bottom-quintile share at Massachusetts public colleges is

+1.38 percentage points ($p < 0.001$), remarkably close to Tennessee's +1.27 pp. Mean parental income rank declines by 1.08 percentage points ($p < 0.001$). The Massachusetts result provides independent validation of the Tennessee finding: a second state with a moderate eligibility threshold and a public-sector subsidy concentration produces a progressive compositional shift of similar magnitude.

8.6 South Dakota Opportunity Scholarship

South Dakota's Opportunity Scholarship requires a minimum ACT score of 24 (approximately the 75th percentile nationally) and provides a comparatively small award ($1,000–$5,000 per year). The high threshold and small subsidy satisfy the model's conditions for regressive sorting: only upper-income students are likely to qualify, and the price differential is too small to induce meaningful sector switching.

The Callaway–Sant'Anna ATT confirms this prediction: bottom-quintile share at South Dakota public colleges *declines* by 1.47 percentage points ($p < 0.001$), and mean parental income rank *increases* by 3.25 percentage points ($p < 0.001$). The effect is strongly regressive—the opposite of Tennessee and Massachusetts. The quintile decomposition is also inverted: Q1 through Q3 lose share, Q4 and Q5 gain share.

8.7 Four-State Comparison

Figure 17 displays event-study estimates for bottom-quintile share across all four states. The visual contrast is striking: Tennessee and Massachusetts trend upward after treatment (progressive), South Dakota trends sharply downward (regressive), and West Virginia shows no clear pattern (null, confounded by enrollment contraction).

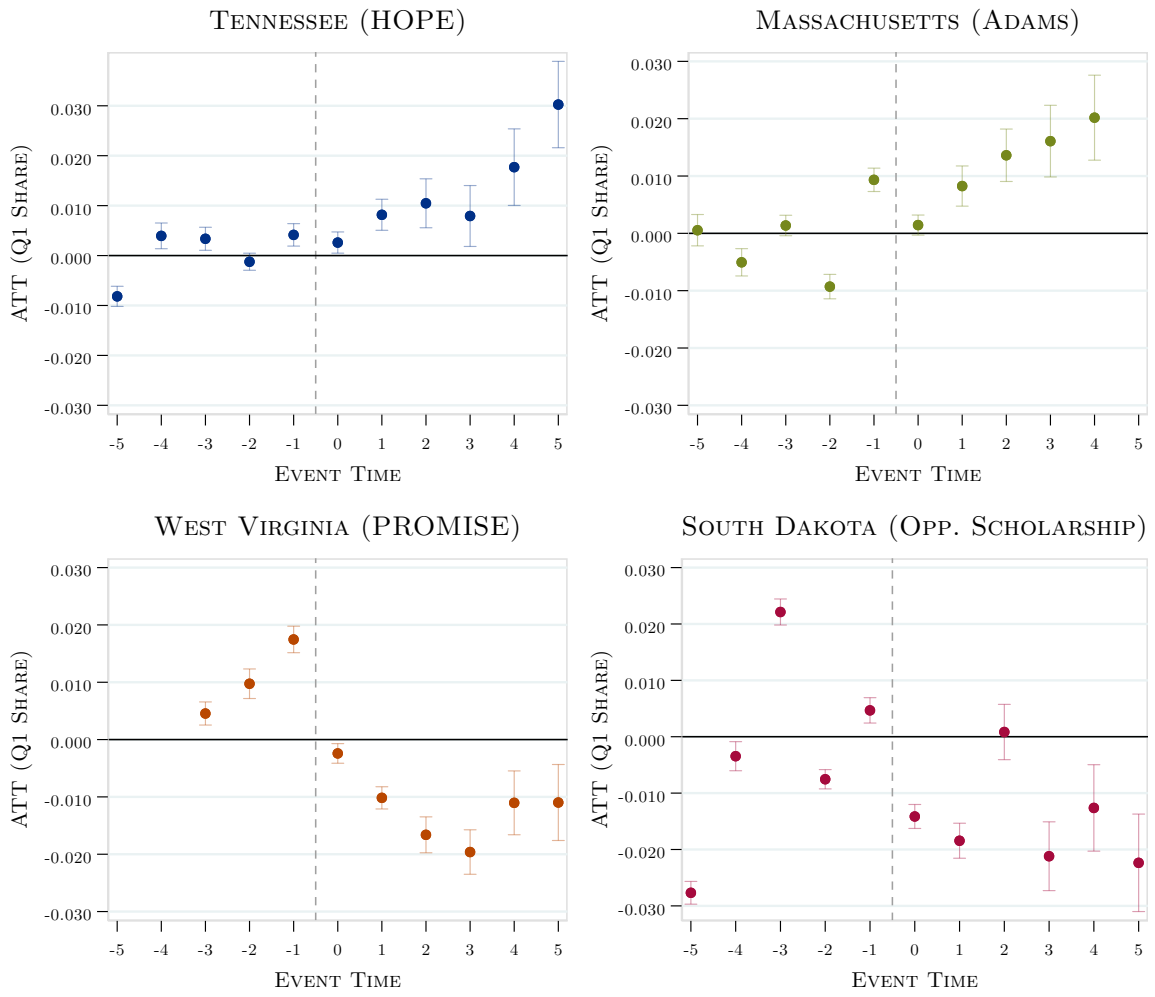

Figure 17: Cross-State Evidence: Merit Aid and Institutional Composition

Notes: Callaway–Sant'Anna event-study estimates for bottom-quintile share at public colleges. Tennessee and Massachusetts show progressive compositional shifts; South Dakota shows a regressive shift; West Virginia shows no significant pattern. Vertical bars denote 95% confidence intervals. Standard errors clustered at the state level. Data: Chetty et al. (2020).

Table 6 maps program design parameters to predicted and observed sorting direction. The three-destination model correctly predicts the direction of compositional change in all four cases where program parameters and institutional context are accounted for.

Table 6: Program Design and Direction of Compositional Sorting

State	Program	Threshold	Public Subsidy	Direction
Tennessee	HOPE	Moderate (3.0 GPA)	Large	Progressive
Massachusetts	Adams	Moderate (top 25%)	Large	Progressive
Georgia	HOPE	Moderate (3.0 GPA)	Muted (Pell offset)	Regressive
South Dakota	Opp. Scholarship	High (ACT 24+)	Small	Regressive
West Virginia	PROMISE	Moderate (3.0 GPA)	Large	Null[†]

Notes: "Threshold" indicates the stringency of academic eligibility. "Public Subsidy" indicates the size and sector-concentration of the scholarship. "Direction" reports the observed sign of the bottom-quintile share effect. [†]West Virginia's null result reflects enrollment contraction rather than a failure of the sorting mechanism (Section 8).

8.8 Generalizability

The four-state comparison extends the evidence beyond a single treated state and addresses the concern that Tennessee's result may be idiosyncratic. Massachusetts independently produces a progressive shift of similar magnitude under similar program parameters, while South Dakota's regressive shift confirms the model's prediction that high thresholds and small subsidies generate the opposite pattern. The three-destination model organizes all five cases (including Georgia's documented regressive pattern) through a common set of design parameters: eligibility threshold location, public-private subsidy differential, and need-based aid interactions.

The WV non-replication highlights an important scope condition: program design is necessary but not sufficient. Enrollment contraction at WV public colleges overwhelmed any compositional dynamics, suggesting that the model's predictions require a stable or growing public system with absorptive capacity for the sorting mechanism to operate.

9 Conclusion

We document that following Tennessee's HOPE Scholarship, public colleges experienced a progressive shift in socioeconomic composition. The share of bottom-quintile students increased by 1.3 percentage points, mean parental income rank declined by 1.3 percentage points, and the effect is

monotonically ordered across all five income quintiles. Lower-income students gained representation while higher-income students lost it.

This compositional change reflects sorting across sectors rather than enrollment expansion. Total enrollment at Tennessee public colleges is unchanged post-HOPE. The level decomposition confirms that the number of bottom-quintile students increased while the number of top-quintile students decreased. Top-quintile students who exit the public sector do not appear at Tennessee private colleges, consistent with out-of-state enrollment rather than in-state public-to-private substitution as the relevant margin.

These results reconcile with the finding that merit aid widens aggregate attendance gaps (Dynarski, 2000). Merit programs can simultaneously increase total college-going among higher-income students (who are more likely to meet eligibility thresholds) and improve the composition of public institutions (because the public sector absorbs disproportionately lower-income new entrants). The three-destination sorting framework—public, private, and out-of-state—resolves what has appeared to be a contradiction in the literature.

The cross-state evidence supports the model's comparative statics. Massachusetts' Adams Scholarship, which shares Tennessee's design features (moderate threshold, public-sector subsidy concentration), independently produces a progressive shift of similar magnitude (+1.38 pp for bottom-quintile share). South Dakota's Opportunity Scholarship, with a high ACT threshold and small award, produces a strongly regressive shift—as the model predicts. West Virginia's PROMISE Scholarship, despite similar design parameters to Tennessee, does not produce progressive sorting because enrollment at WV publics contracted during the analysis period. These results suggest that program design determines the *direction* of sorting, while institutional context determines whether sorting *operates*.

Four limitations merit emphasis. First, the analysis relies on ecological inference from college-level aggregates. While the level decomposition addresses the most direct concern, we cannot observe individual enrollment decisions or track which students switch sectors. Second, with 27 state-level clusters, inference is at the boundary of reliable asymptotic approximation; we address this with permutation inference. Third, the Δ_{SD} sensitivity analysis of Rambachan and Roth (2023) indicates that results are fragile under smoothness restrictions on differential trends. We interpret this as a genuine bound on what the research design can deliver, not as evidence against

the effect. Fourth, the WV non-replication raises the question of whether Tennessee's progressive result generalizes or reflects state-specific conditions.

For policy, the findings suggest that the distributional consequences of merit aid depend critically on program design. When eligibility thresholds are set low enough that a substantial share of lower-income students qualify, and when the subsidy is concentrated at public institutions, merit aid can function as a progressive sorting mechanism. The relevant counterfactual is not whether merit aid is more progressive than need-based aid—it almost certainly is not—but whether merit aid *at the margin* improves or harms the composition of the institutions that serve the most students. If states want merit programs to improve access, the evidence suggests that the threshold should be moderate (qualifying a broad share of the income distribution, not only top students), the subsidy should be large enough to affect enrollment decisions (not a token amount), and the institutional system should be large enough to absorb compositional shifts without enrollment contraction.

Descriptive evidence on upward mobility rates (Section 8.3) suggests that the progressive composition shift did not come at the cost of diminished returns: Tennessee public colleges' mobility rates converged toward the national average in the post-HOPE period.

Several directions for future research follow from these findings. First, a structural model using individual-level administrative data could identify the specific enrollment decisions—access expansion, sector switching, out-of-state exit—that drive the aggregate compositional shifts we document. Second, additional state cases with large, stable public systems and clean program variation would test the model's predictions more powerfully than the WV case, where enrollment contraction confounds identification. Third, longer post-treatment horizons would reveal whether the progressive shift persists or attenuates as institutions adjust capacity and admissions standards. Fourth, spatial spillover effects on neighboring states—if Tennessee retains students who would otherwise cross borders—deserve investigation as a potential general equilibrium channel.

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A Figures

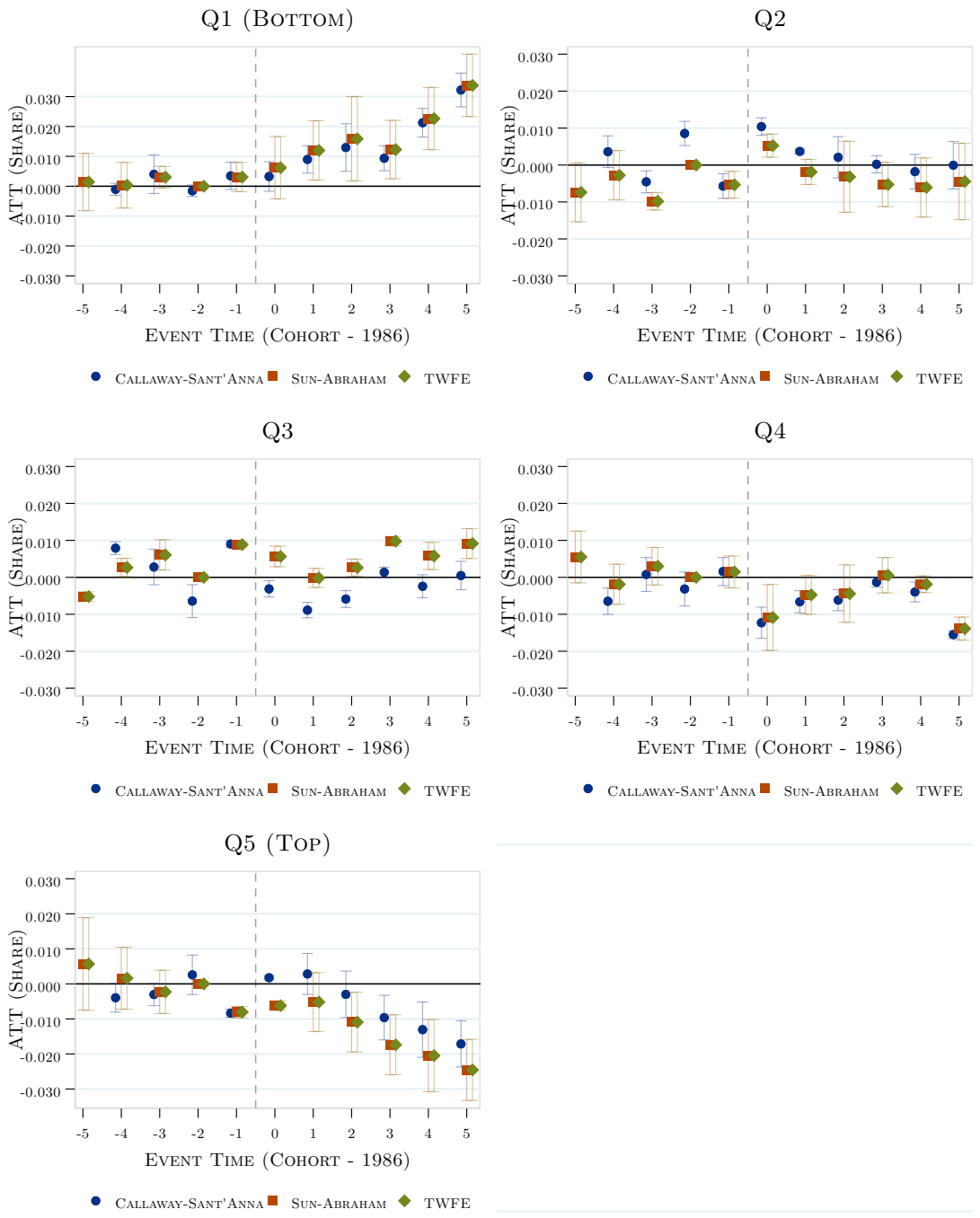

Figure A1: Event-Study Estimates by Income Quintile

Notes: Event-study estimates from three heterogeneity-robust estimators (Callaway & Sant'Anna (2021), Sun & Abraham (2021), and TWFE), estimated separately for each income quintile share. Bottom quintiles (Q1–Q2) trend upward post-treatment (gaining share); top quintiles (Q4–Q5) trend downward (losing share); Q3 is near zero. The monotonic gradient is consistent across all three estimators. Bordering-state control group (AL, NC, VA). $k = -2$ omitted (1984 cohort, last untreated). 1980 birth cohort dropped. Standard errors clustered by state. Data: Chetty et al. (2020).

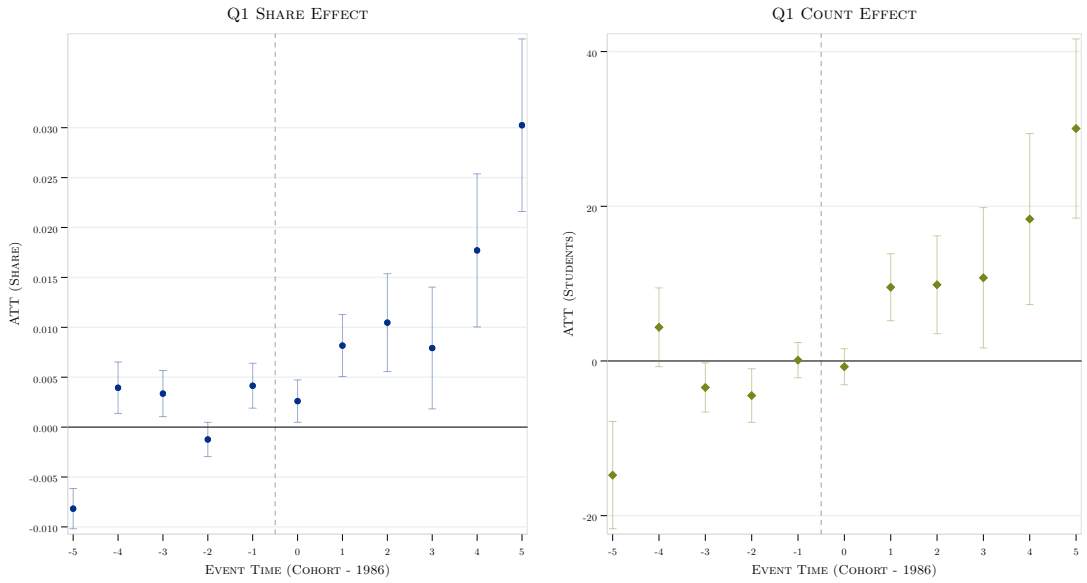

Figure A2: Share Effects vs. Count Effects: Bottom-Quintile Students

Notes: Comparison of share-based (left) and count-based (right) event-study estimates for bottom-quintile students. The share increase is accompanied by a count increase, confirming that the composition shift reflects genuine sorting rather than a denominator artifact. Count outcomes are inherently noisier than share outcomes because they inherit enrollment variation across colleges (coefficient of variation ≈ 1.0). The wider confidence intervals on count estimates reflect this additional noise, not weaker identification. Callaway & Sant'Anna (2021) estimator. Data: Chetty et al. (2020).

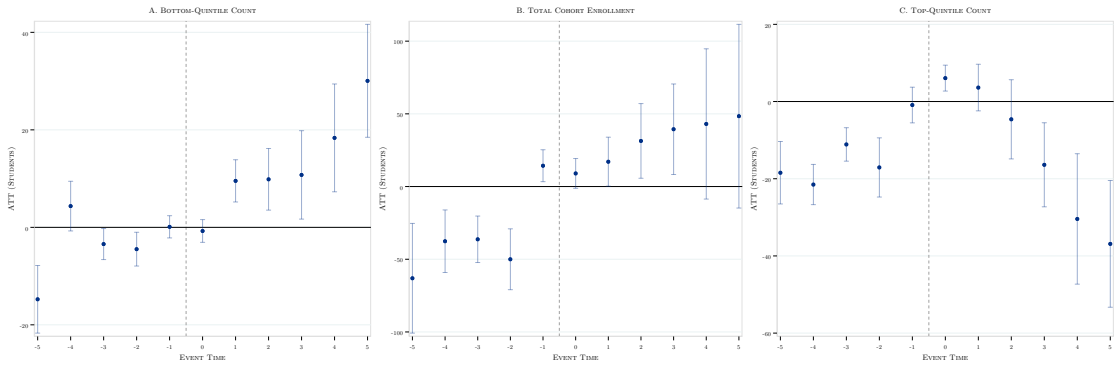

Figure A3: Level Decomposition: Bottom-Quintile Count, Top-Quintile Count, and Total Enrollment

Notes: Callaway & Sant'Anna (2021) event-study estimates for count outcomes (quintile share × cohort enrollment). The critical finding is that total enrollment (Panel B) is unchanged post-treatment, ruling out the denominator-artifact explanation for share-based results. Q1 count increases and Q5 count decreases, consistent with real compositional sorting. Count outcomes exhibit wider confidence intervals than share outcomes because they compound share-level variation with enrollment variation (enrollment coefficient of variation ≈ 1.0 across colleges). The pre-treatment noise in count event studies reflects this inherent volatility rather than a failure of parallel trends for the share-based specifications. Data: Chetty et al. (2020).

Figure A4: Spillover Test: Tennessee Private Colleges

Notes: Callaway & Sant'Anna (2021) event-study estimates comparing Tennessee private colleges (treated) to private colleges in control states. No significant composition change is detected, indicating that top-quintile students who exit Tennessee public colleges do not re-sort to in-state private institutions. $N = 5{,}336$. Data: Chetty et al. (2020).

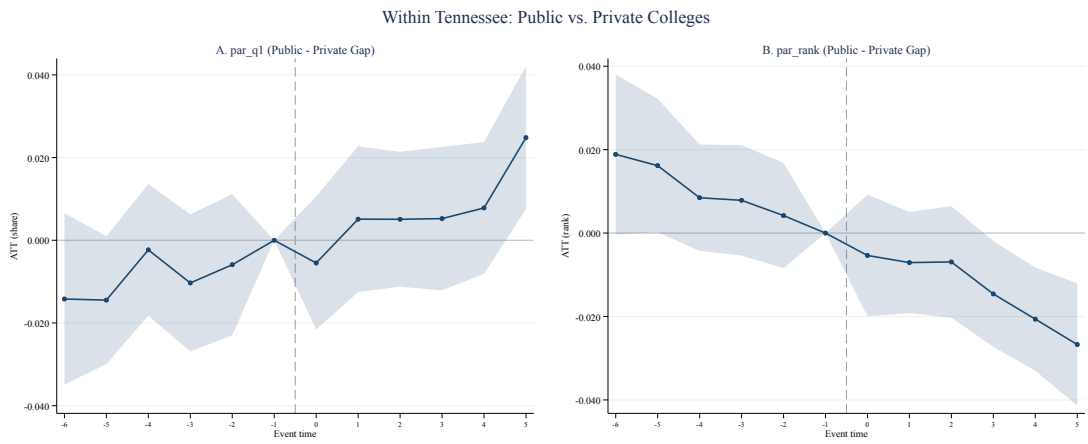

Figure A5: Within-Tennessee Comparison: Public vs. Private Colleges

Notes: Event-study estimates restricting the sample to Tennessee institutions only. Public colleges (treated) are compared to Tennessee private colleges (control). The composition shift is concentrated in the public sector. College and cohort fixed effects included. $N = 571$. Data: Chetty et al. (2020).

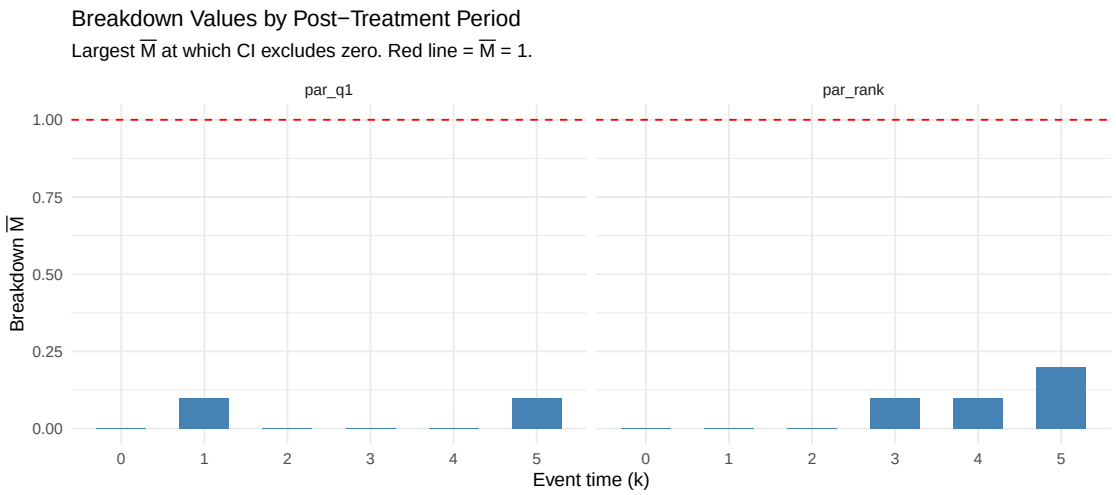

Figure A6: Sensitivity Analysis: Breakdown Values Across Post-Treatment Event Times

Notes: Breakdown values of $\bar{M}$ at each post-treatment event time under both the relative magnitude (Δ_{RM}) and smoothness (Δ_{SD}) restrictions of Rambachan & Roth (2023). Higher breakdown values indicate greater robustness to violations of parallel trends. Data: Chetty et al. (2020).

B Tables

Table A1: Distribution of HOPE Scholarship Recipients by Sector, Fall 2004–2008

	2004	2005	2006	2007	2008
Panel A: Number of Recipients					
TBR Universities	7,454	7,054	7,637	7,956	8,121
TBR Community Colleges	4,506	3,909	4,720	4,996	4,970
UT Campuses	5,380	5,310	5,576	5,988	6,327
Private Colleges	3,109	3,387	3,492	3,885	3,902
Total	**20,449**	**19,660**	**21,425**	**22,825**	**23,320**
Panel B: Share of Recipients (%)					
TBR Universities	36	36	36	35	35
TBR Community Colleges	22	20	22	22	21
UT Campuses	26	27	26	26	27
Private Colleges	15	17	16	17	17

Notes: First-time freshmen receiving Tennessee HOPE Scholarship awards. TBR = Tennessee Board of Regents; UT = University of Tennessee system. Panel B shares may not sum to 100 due to rounding. Public institutions (TBR + UT) account for 83–85% of recipients in all years. Source: Tennessee Student Assistance Corporation (Tennessee Higher Education Commission, 2025).

Table A2: Difference-in-Differences: Pre- and Post-Treatment Means

	Treated (TN Public)	**Control** (Non-Merit States)
Panel A: Pre-Treatment Means		
Bottom-quintile share	0.138	0.120
Panel B: Post-Treatment Means		
Bottom-quintile share	0.159	0.129
Panel C: DiD Estimate		
Treated × Post	0.0148^{***}	
	(0.0016)	
College FE	✓	
Cohort FE	✓	
Observations	23,395	
R^2	0.948	
Clusters	27	

Notes: Standard errors clustered at the state level in parentheses. Treated = Tennessee public colleges. Control = public colleges in 26 states without broad-based merit programs (Sjoquist & Winters, 2015). Post = birth cohorts 1985 and later. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A3: Callaway–Sant'Anna Aggregate ATT Estimates

	(1) Bottom-Quintile Share	(2) Parental Income Rank
ATT	0.0127^{***}	-0.0133^{***}
	(0.0026)	(0.0020)
95% CI	[0.0075, 0.0179]	[−0.0172, −0.0094]

Notes: Aggregate average treatment effect on the treated from Callaway & Sant'Anna (2021). The comparison group consists of never-treated public colleges in 26 non-merit-program states. Treatment onset: $t^* = 1986$. Analytical standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A4: Static TWFE Difference-in-Differences Estimates

	(1) Bottom-Quintile Share	(2) Parental Income Rank
TN Public × Post	0.0154^{***}	-0.0185^{***}
	(0.0030)	(0.0025)
Constant	0.138^{***}	0.540^{***}
	(0.0001)	(0.0001)
Observations	6,289	6,289
Adjusted R^2	0.935	0.974
Clusters	27	27

Notes: Two-way fixed effects estimates with college and cohort fixed effects. Standard errors clustered at the state level in parentheses. Treated = Tennessee public colleges. Post = birth cohort ≥ 1986. The sample includes public colleges in Tennessee and 26 non-merit-program states. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A5: Quintile Decomposition: TWFE Static DiD by Income Quintile

	(1)	(2)	(3)	(4)	(5)
	Q1 (Bottom)	Q2	Q3	Q4	Q5 (Top)
TN Public × Post	0.0154***	0.0114***	0.0047*	−0.0136***	−0.0179***
	(0.0030)	(0.0025)	(0.0018)	(0.0021)	(0.0025)
Observations	6,289	6,289	6,289	6,289	6,289

Notes: Each column reports the TWFE static difference-in-differences estimate from a separate regression in which the dependent variable is the share of students from the indicated income quintile. College and cohort fixed effects included. Standard errors clustered at the state level in parentheses (27 clusters). Quintile shares sum to one by construction. Post = birth cohort $\geq$ 1986. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A6: Level Decomposition: Share and Count Outcomes

	(1)	(2)	(3)	(4)	(5)
	Q1 Share	Q1 Count	Total Count	Q5 Share	Q5 Count
TN Public × Post	0.0154***	11.23**	−2.11	−0.0179***	−34.38***
	(0.0030)	(4.256)	(25.42)	(0.0025)	(8.635)
Constant	0.138***	167.9***	1,586***	0.200***	449.7***
	(0.0001)	(0.104)	(0.622)	(0.0001)	(0.211)
Observations	6,289	6,289	6,289	6,289	6,289
Adjusted R^2	0.935	0.948	0.985	0.978	0.989
Clusters	27	27	27	27	27

Notes: Two-way fixed effects estimates with college and cohort fixed effects. Standard errors clustered at the state level in parentheses. Count outcomes are constructed as quintile share × cohort size. The Q1 share increase (column 1) is accompanied by a Q1 count increase (column 2), while total enrollment is unchanged (column 3), confirming that share changes reflect compositional sorting rather than a denominator artifact. Post = birth cohort $\geq$ 1986. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A7: Heterogeneity by Institution Tier: Two-Year vs. Four-Year Colleges

	Two-Year			Four-Year		
Outcome	Coef.	SE	N	Coef.	SE	N
Bottom-quintile share	0.0181^{***}	(0.0034)	4,016	0.0070^{**}	(0.0024)	2,259
Parental income rank	-0.0173^{***}	(0.0028)	4,016	-0.0186^{***}	(0.0021)	2,259
Q1 count	5.73	(5.56)	4,016	19.60^{**}	(5.53)	2,259
Q5 count	-22.94^{**}	(6.24)	4,016	-39.23^{*}	(17.55)	2,259
Total count	-58.41	(35.54)	4,016	138.52^{***}	(34.66)	2,259

Notes: TWFE static difference-in-differences estimates estimated separately for two-year (community colleges) and four-year institutions. College and cohort fixed effects included. Standard errors clustered at the state level in parentheses. Two-year colleges show displacement (composition shifts, enrollment unchanged); four-year colleges show expansion (enrollment grows, composition shifts). Post = birth cohort $\geq$ 1986. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A8: Spillover Test: Tennessee Private Colleges

Outcome	Coef.	SE	p-value	N
Bottom-quintile share	0.0011	(0.0011)	0.305	5,336
Parental income rank	-0.0046^{*}	(0.0018)	0.016	5,336

Notes: Callaway & Sant'Anna (2021) aggregate ATT comparing Tennessee private colleges to private colleges in 26 non-merit-program control states. The null effect on bottom-quintile share indicates that top-quintile students exiting Tennessee public colleges do not re-sort to in-state private institutions. Analytical standard errors in parentheses. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A9: Within-Tennessee Comparison: Public vs. Private Colleges

Outcome	Coef.	SE	p-value	N
Bottom-quintile share	0.0142^{***}	(0.0037)	< 0.001	571
Parental income rank	-0.0214^{***}	(0.0032)	< 0.001	571

Notes: TWFE static difference-in-differences restricted to Tennessee institutions only. Treated = Tennessee public colleges; control = Tennessee private colleges. College and cohort fixed effects included. Standard errors in parentheses. The small sample reflects the within-state restriction ($N = 571$). Post = birth cohort $\geq$ 1986. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).

Table A10: Robustness Summary Across Estimators, Timing, and Control Groups

	Bottom-Quintile Share		Parental Income Rank	
Specification	Coef.	SE	Coef.	SE
Panel A: Estimator				
C&S (preferred)	0.0127^{***}	(0.0026)	-0.0133^{***}	(0.0020)
TWFE static	0.0154^{***}	(0.0030)	-0.0185^{***}	(0.0025)
Panel B: Treatment Timing				
$t^* = 1985$	0.0144^{***}	(0.0028)	-0.0180^{***}	(0.0024)
$t^* = 1986$ (preferred)	0.0154^{***}	(0.0030)	-0.0185^{***}	(0.0025)
$t^* = 1987$	0.0164^{***}	(0.0031)	-0.0178^{***}	(0.0025)
Panel C: Control Group				
Baseline (26 states)	0.0154^{***}	(0.0030)	-0.0185^{***}	(0.0025)
Conservative	0.0151^{***}	(0.0032)	-0.0175^{***}	(0.0024)
Bordering (preferred)	0.0144^{***}	(0.0034)	-0.0134^{**}	(0.0049)
Panel D: Sensitivity				
Δ_{RM}: breakdown $\bar{M}$	2		2	
Δ_{SD}: includes zero at	moderate $\bar{M}$		moderate $\bar{M}$	

Notes: Panels A–C report static difference-in-differences estimates under alternative specifications. All specifications include college and cohort fixed effects. Standard errors clustered at the state level in parentheses. Panel D summarizes Rambachan & Roth (2023) sensitivity results: the Δ_{RM} breakdown value indicates the maximum $\bar{M}$ at which the confidence interval excludes zero; the Δ_{SD} row indicates that confidence intervals include zero at moderate values of $\bar{M}$. $^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$. Data: Chetty et al. (2020).